

# QUADRATIC EQUATIONS & EXPRESSIONS



## SYNOPSIS

- **Quadratic Expression:** If  $a \neq 0$ ,  $b, c$  are real (or) complex numbers then  $ax^2 + bx + c$  is called a quadratic expression in  $x$   
**Ex:**  $x^2 - 7x + 12$ ,  $x^2 + 8x + 12$ ,  $x^2 - (1 + 2i)x + 38$ .
- A complex number ' $\alpha$ ' is said to be a 'zero' of quadratic expression  $ax^2 + bx + c$ , if  $a\alpha^2 + b\alpha + c = 0$   
**Ex:** 3 is a zero of  $x^2 - 5x + 6$
- **Quadratic Equation:** If  $a \neq 0$ ,  $b, c$  are real (or) complex numbers then  $ax^2 + bx + c = 0$  is called a quadratic equation in  $x$ .  
**Ex:**  $x^2 + 5x + 6 = 0$ ,  $x^2 + x + 1 = 0$
- **Identity :** A relation which is true for every value of the variable is called an identity
- **Quadratic Identity:**  $ax^2 + bx + c = 0$  will be an identity (or can have more than two solutions) if  $a = 0, b = 0, c = 0$ .
- **Root of a Quadratic Equation:**  
 If  $a\alpha^2 + b\alpha + c = 0$  then  $\alpha$  is a root or solution of the quadratic equation  $ax^2 + bx + c = 0$ .
- The roots of the quadratic equation  $ax^2 + bx + c = 0$  are  $\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$  and its discriminant is  $\Delta = b^2 - 4ac$ .

**W.E-1:** The roots of  $3x^2 - 5x - 12 = 0$  are

**Sol:**  $x = \frac{5 \pm \sqrt{25 - 4(3)(-12)}}{2 \times 3}$  i.e.  $x = -\frac{4}{3}, 3$

or (using factorization method)

$$3x^2 - 9x + 4x - 12 = 0$$

$$3x(x - 3) + 4(x - 3) = 0$$

$$(3x + 4)(x - 3) = 0 ; x = -\frac{4}{3}, 3$$

- If  $\alpha$  and  $\beta$  are the roots of  $ax^2 + bx + c = 0$  then,
  - i)  $\alpha + \beta = -\frac{b}{a}, \alpha\beta = \frac{c}{a}$

$$\text{ii) } |\alpha - \beta| = \frac{\sqrt{b^2 - 4ac}}{|a|}$$

$$\text{iii) } \alpha^2 + \beta^2 = \frac{b^2 - 2ac}{a^2}$$

$$\text{iv) } \alpha^3 + \beta^3 = \frac{3abc - b^3}{a^3}$$

- If  $\alpha$  and  $\beta$  are the roots of  $ax^2 + bx + c = 0$  then the quadratic expression  $ax^2 + bx + c = a(x - \alpha)(x - \beta)$
- The quadratic equation whose roots are  $\alpha, \beta$  is given by  $x^2 - (\alpha + \beta)x + \alpha\beta = 0$

**Note:** Quadratic equation cannot have more than two roots.

**W.E-2 :** The quadratic equation whose roots are

$$1 + \sqrt{2}, 1 - \sqrt{2} \text{ is}$$

**Sol:**  $x^2 - (1 + \sqrt{2} + 1 - \sqrt{2})x + (1 + \sqrt{2})(1 - \sqrt{2}) = 0$

$$\text{i.e. } x^2 - 2x + (-1) = 0$$

$$x^2 - 2x - 1 = 0 \text{ (or) } x = 1 + \sqrt{2}$$

$$x - 1 = \sqrt{2} \quad (x - 1)^2 = 2$$

$$x^2 - 2x - 1 = 0$$

**W.E-3 :** The quadratic equation whose roots are

$$\frac{3 \pm i\sqrt{5}}{2} \text{ is}$$

**Sol:** 
$$x^2 - \left( \frac{3+i\sqrt{5}}{2} + \frac{3-i\sqrt{5}}{2} \right)x + \left( \frac{3+i\sqrt{5}}{2} \cdot \frac{3-i\sqrt{5}}{2} \right) = 0$$

$$x^2 - 3x + \frac{7}{2} = 0 \Rightarrow 2x^2 - 6x + 7 = 0$$

$$\text{or } x = \frac{3+i\sqrt{5}}{2} \Rightarrow (2x-3)^2 = -5$$

$$\Rightarrow 2x^2 - 6x + 7 = 0$$

**W.E-4:** If  $\alpha, \beta$  are the roots of  $ax^2 + bx + c = 0$  and  $c \neq 0$  then the value of

$$\frac{1}{(a\alpha + b)^2} + \frac{1}{(a\beta + b)^2} \text{ in terms of } a, b, c \text{ is}$$

**Sol:**  $\because \alpha$  is a root of  $ax^2 + bx + c = 0$

$$\Rightarrow a\alpha^2 + b\alpha = -c$$

$$\Rightarrow a\alpha + b = \frac{-c}{\alpha}, \text{ similarly } a\beta + b = \frac{-c}{\beta}$$

$$\begin{aligned} \therefore \frac{1}{(a\alpha + b)^2} + \frac{1}{(a\beta + b)^2} \\ = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{c^2} = \frac{b^2 - 2ac}{a^2c^2} \end{aligned}$$

**Nature of the roots of the equation**

$$ax^2 + bx + c = 0 :$$

➤ If  $a, b, c$  are real and

i)  $\Delta > 0$ , then the roots are real and distinct

ii)  $\Delta = 0$ , then the roots are real and equal

iii)  $\Delta < 0$ , then the roots are non-real conjugate complex numbers i.e.,  $\alpha \pm i\beta$

➤ If  $a, b, c$  are rational and

i)  $\Delta > 0$  and is a perfect square then the roots are rational and distinct.

ii)  $\Delta > 0$  and is not a perfect square then the roots are conjugate surds i.e.,  $\alpha \pm \sqrt{\beta} \cdot (\beta \neq 0)$

iii)  $\Delta = 0$ , then the roots are equal & rational

iv)  $\Delta < 0$ , then the roots are non-real conjugate complex numbers. i.e.,  $\alpha \pm i\beta$

**W.E-5:** The nature of the roots of  $x^2 + x - 1 = 0$  is

**Sol:**  $\Delta = (1)^2 - 4(1)(-1) = 5$  is not a perfect square.  
 $\Rightarrow$  roots are irrational and more over conjugate

**W.E-6:** The nature of the roots of  $2x^2 + x + 6 = 0$  is

**Sol:**  $\Delta = 1 - 4(2)(6) = -47 < 0$   
 $\Rightarrow$  roots are imaginary more over conjugate complex numbers

**W.E-7:** If the roots of the equation  $x^2 - 15 - m(2x - 8) = 0$  are equal then the value of  $m$  is

**Sol:**  $x^2 - 15 - m(2x - 8) = 0$

$$x^2 - 2mx + 8m - 15 = 0$$

$$\Delta = 0 \Rightarrow (-2m)^2 - 4 \times 1 \times (8m - 15) = 0$$

$$m^2 - 8m + 15 = 0$$

$$(m-3)(m-5) = 0 \quad m = 3; m = 5$$

**W.E-8:** If  $a, b, c$  are real, the roots of

$(a^2 + b^2)x^2 - 2b(a+c)x + b^2 + c^2 = 0$  are real and equal, then the progression in which  $a, b, c$  lies is

**Sol:**  $a^2x^2 + b^2x^2 - 2abx - 2bcx + b^2 + c^2 = 0$

$$a^2x^2 - 2abx + b^2 + b^2x^2 - 2bcx + c^2 = 0$$

$$(ax - b)^2 + (bx - c)^2 = 0$$

$$ax - b = 0; \quad bx - c = 0$$

$$x = \frac{b}{a}; \quad x = \frac{c}{b}$$

$$\frac{b}{a} = \frac{c}{b} \Rightarrow b^2 = ac, \therefore a, b, c \text{ are in G.P}$$

**W.E-9 :** If the roots of the equation

$$(a^2 - bc)x^2 + 2(b^2 - ca)x + (c^2 - ab) = 0$$

are equal then the condition(s) is (are)

**Sol:**  $(a^2 - bc)x^2 + 2(b^2 - ca)x + (c^2 - ab) = 0$

roots are equal  $\Rightarrow \Delta = 0$

$$\Rightarrow 4(b^2 - ca)^2 - 4(a^2 - bc)(c^2 - ab) = 0$$

$$\Rightarrow b^4 - 2ab^2c + a^3b + bc^3 - ab^2c = 0$$

$$\Rightarrow b = 0 \text{ (or) } a^3 + b^3 + c^3 - 3abc = 0$$

➤ **Nature of the roots of two Quadratic Equations:** If  $\Delta_1$  and  $\Delta_2$  are the discriminants of two quadratic equations  $P(x) = 0$  and  $Q(x) = 0$ , such that

➤  $\Delta_1 + \Delta_2 \geq 0$  then there will be at least two real roots for the equation  $P(x) = 0$  or  $Q(x) = 0$ .

➤ If  $\Delta_1 + \Delta_2 < 0$ , then there will be atleast two imaginary roots for the equation  $P(x) = 0$  or  $Q(x) = 0$ .

➤ If  $\Delta_1 \cdot \Delta_2 < 0$ , then the equation  $P(x) \cdot Q(x) = 0$  will have two real roots and two imaginary roots.

➤ If  $\Delta_1 \cdot \Delta_2 > 0$ , then the equation  $P(x) \cdot Q(x) = 0$  has either four real roots or no real roots.

➤ If  $\Delta_1 \cdot \Delta_2 = 0$  such that  $\Delta_1 > 0$  and  $\Delta_2 = 0$  or  $\Delta_1 = 0$  and  $\Delta_2 > 0$  then the equation  $P(x) \cdot Q(x) = 0$  will have two equal roots and two distinct roots.

➤ If  $\Delta_1 \cdot \Delta_2 = 0$  where  $\Delta_1 < 0$  and  $\Delta_2 = 0$  or  $\Delta_1 = 0$  and  $\Delta_2 < 0$  then the equation  $P(x) \cdot Q(x) = 0$  will have two equal real roots.

➤ If  $\Delta_1 \cdot \Delta_2 = 0$  such that  $\Delta_1 = 0$  and  $\Delta_2 = 0$  then the equation  $P(x) \cdot Q(x) = 0$  will have two pairs of equal roots.

➤ **Common root:** The equations  $a_1x^2 + b_1x + c_1 = 0$  and  $a_2x^2 + b_2x + c_2 = 0$  where  $a_1b_2 - a_2b_1 \neq 0$ ,  $a_1a_2 \neq 0$  have a common root if and only if

$$(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1) \text{ and}$$

the common root is  $\frac{c_1a_2 - c_2a_1}{a_1b_2 - a_2b_1}$  (or)  $\frac{b_1c_2 - b_2c_1}{c_1a_2 - c_2a_1}$

**W.E-10 :** If  $x^2 + 4ax + 3 = 0$  and  $2x^2 + 3ax - 9 = 0$  have a common root then the value of  $a$  is

**Sol:** Condition for the common roots is

$$(c_1a_2 - c_2a_1)^2 = (a_1b_2 - a_2b_1)(b_1c_2 - b_2c_1)$$

$$(6+9)^2 = (3a-8a)(-36a-9a)$$

$$\Rightarrow a = \pm 1$$

**W.E-11:** If  $x^2 + bx + c = 0$ ,  $x^2 + cx + b = 0$

( $b \neq c$ ), have a common root then the condition is

**Sol:** Let  $\alpha$  be the common root

$$\Rightarrow \alpha^2 + b\alpha + c = 0 \dots (1)$$

$$\alpha^2 + c\alpha + b = 0 \dots (2)$$

$$(1) - (2) \Rightarrow \alpha = 1$$

$$\text{substitute in (1)} \Rightarrow b + c + 1 = 0$$

**Note:** If  $a_1x^2 + b_1x + c_1 = 0$  and  $a_2x^2 + b_2x + c_2 = 0$

$$\text{have the same roots then } \frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

**W.E-12:** If the equation

$$k(6x^2 + 3) + rx + (2x^2 - 1) = 0 \text{ and}$$

$$6k(2x^2 + 1) + px + (4x^2 - 2) = 0 \text{ have both roots common, then the value of } p/r \text{ is}$$

**Sol:**  $(6k + 2)x^2 + rx + 3k - 1 = 0 \rightarrow (1)$

$$(12k + 4)x^2 + px + 6k - 2 = 0 \rightarrow (2)$$

since the roots of (1) and (2) are equal,

$$\frac{6k+2}{12k+4} = \frac{r}{p} = \frac{3k-1}{6k-2} \therefore \frac{r}{p} = \frac{1}{2}$$

➤ **Properties of roots of the equation**

$ax^2 + bx + c = 0$  : If  $a$  and  $c$  are of the same sign

then  $\frac{c}{a}$  is +ve and hence roots have same signs

➤ If  $a$  and  $c$  are of the opposite signs then  $\frac{c}{a}$  is -ve and hence roots have opposite sign

➤ If  $a = c$  then the roots are reciprocal to each other.

➤ If both the roots are -ve then  $a, b, c$  will have the same sign

- If both roots are +ve then a, c will have the same sign different from the sign of b
- If  $a+b+c=0$  then the roots are 1 and  $\frac{c}{a}$
- If  $a+c=b$  then the roots are -1 and  $\frac{-c}{a}$
- If the roots are in the ratio  $m : n$  then
 
$$(m+n)^2 ac = mn b^2 \text{ (or) } \frac{(m+n)^2}{mn} = \frac{b^2}{ac}$$
- If one root is square of the other then
 
$$a^2 c + c^2 a = b(3ac - b^2)$$
- If one root is equal to the  $n^{\text{th}}$  power of the other root then
 
$$(ac^n)^{\frac{1}{n+1}} + (a^n c)^{\frac{1}{n+1}} + b = 0$$
- If roots differ by k then  $b^2 - 4ac = a^2 k^2$ .

**W. E-13: The roots of the equation**

$$a(b-c)x^2 + b(c-a)x + c(a-b) = 0$$

$(a \neq b \neq c \neq 0)$  are

**Sol:**  $a(b-c) + b(c-a) + c(a-b) = 0$

$$\therefore \text{the roots are } 1, \frac{c(a-b)}{a(b-c)}$$

**W.E-14: If  $p(q-r)x^2 + q(r-p)x + r(p-q) = 0$**

$(p \neq q \neq r)$  has equal roots the value of  $\frac{2}{q}$  in terms of p & q is

**Sol:**  $p(q-r) + q(r-p) + r(p-q) = 0$

$$\therefore \text{roots are } 1, \frac{r(p-q)}{p(q-r)}$$

$$\therefore \text{roots are equal} \Rightarrow 1 = \frac{r(p-q)}{p(q-r)}$$

$$p(q-r) = r(p-q) \Rightarrow pq - pr = rp - rq$$

$$pq + rq = 2pr \Rightarrow (p+r)q = 2pr$$

$$\frac{p+r}{pr} = \frac{2}{q}$$

**W.E-15: The roots of  $ax^2 + 3bx + c = 0$  if  $3b = a + c$  are**

**Sol:**  $ax^2 + 3bx + c = 0$  and  $3b = a + c$

$$ax^2 + (a+c)x + c = 0; \quad ax^2 + ax + cx + c = 0$$

$$ax(x+1) + c(x+1) = 0; \quad (x+1)(ax+c) = 0$$

$$x = -1; \quad x = \frac{-c}{a} \therefore \text{roots are } -1, \frac{-c}{a}$$

**W.E-16: If one root of the equation  $ax^2 + bx + c = 0$  is double the other, then the relation between a, b, c is**

**Sol:** Condition is  $\frac{(m+n)^2}{mn} = \frac{b^2}{ac}$  given  $m : n = 1 : 2$

$$\therefore \frac{(1+2)^2}{1 \times 2} = \frac{b^2}{ac} \Rightarrow 9ac = 2b^2$$

- **Transformed Equations:** Let  $\alpha, \beta$  be the roots of  $f(x) = ax^2 + bx + c = 0$  then

| S.No | New Roots                                        | Corresponding Quadratic Equation  |
|------|--------------------------------------------------|-----------------------------------|
| 1    | $-\alpha, -\beta$                                | $f(-x) = 0$                       |
| 2    | $\frac{1}{\alpha}, \frac{1}{\beta}$              | $f\left(\frac{1}{x}\right) = 0$   |
| 3    | $k\alpha, k\beta \ (k \neq 0)$                   | $f\left(\frac{x}{k}\right) = 0$   |
| 4    | $\frac{\alpha}{k}, \frac{\beta}{k} \ (k \neq 0)$ | $f(kx) = 0$                       |
| 5    | $\alpha + k, \beta + k$                          | $f(x - k) = 0$                    |
| 6    | $\alpha^2, \beta^2$                              | $f(\sqrt{x}) = 0$                 |
| 7    | $\alpha^3, \beta^3$                              | $f(\sqrt[3]{x}) = 0$              |
| 8    | $\frac{\alpha}{1+\alpha}, \frac{\beta}{1+\beta}$ | $f\left(\frac{x}{1-x}\right) = 0$ |

**W.E-17: If  $\alpha, \beta$  are the roots of  $ax^2 + bx + c = 0$ , then the equation whose roots are  $2+\alpha$  and  $2+\beta$  is**

**Sol:**  $f(x-2) = 0 \Rightarrow a(x-2)^2 + b(x-2) + c = 0$   
 $\Rightarrow ax^2 + x(b-4a) + 4a - 2b + c = 0$

**W.E-18:** If each root of the equation  $x^2 + 11x + 13 = 0$  is diminished by 4, then the transformed equation is

**Sol:**  $f(x+4) = 0 \Rightarrow (x+4)^2 + 11(x+4) + 13 = 0$   
 $\Rightarrow x^2 + 19x + 73 = 0$

**W.E-19:** If  $\alpha, \beta$  are the roots of  $4x^2 + 7x + 2 = 0$ , then the equation whose roots are  $\alpha^2, \beta^2$  is

**Sol:**  $f(x) = 4x^2 + 7x + 2 = 0$   
 The required equation is  $f(\sqrt{x}) = 0$ .  
 $\Rightarrow 4(\sqrt{x})^2 + 7\sqrt{x} + 2 = 0 \Rightarrow 4x + 2 = -7\sqrt{x}$   
 $\Rightarrow (4x + 2)^2 = (-7\sqrt{x})^2 \Rightarrow 16x^2 - 33x + 4 = 0$

➤ **Signs of 'a' and  $ax^2 + bx + c$ :** If the equation  $ax^2 + bx + c = 0$  has non real roots ( $\Delta < 0$ ) then 'a' and  $ax^2 + bx + c$  will have same sign  $\forall x \in R$ .

➤ If the equation  $ax^2 + bx + c = 0$  has equal roots then 'a' and  $ax^2 + bx + c$  will have same sign  $\forall x \in R - \left\{ \frac{-b}{2a} \right\}$ .

➤ If the equation  $ax^2 + bx + c = 0$  has real roots  $\alpha$  and  $\beta$  ( $\Delta > 0, \alpha < \beta$ ) then

(i)  $\alpha < x < \beta \Leftrightarrow a$  and  $ax^2 + bx + c$  will have opposite sign.

(ii)  $x < \alpha$  or  $x > \beta \Leftrightarrow a$  and  $ax^2 + bx + c$  will have same sign.

➤ **Maximum And Minimum Values of  $ax^2 + bx + c$ ;  $a, b, c \in R$ :**

(i) If  $a > 0$  then  $f(x) = ax^2 + bx + c$  has absolute minimum at  $X = -\frac{b}{2a}$  and the minimum value is

$$\frac{4ac - b^2}{4a}$$

(ii) If  $a < 0$  then  $f(x) = ax^2 + bx + c$  has absolute maximum at  $X = -\frac{b}{2a}$  and the maximum value

$$\text{is } \frac{4ac - b^2}{4a}$$

(iii) If  $f(x) = \frac{ax^2 + bx + c}{ax^2 - bx + c}$  (or)  $f(x) = \frac{ax^2 - bx + c}{ax^2 + bx + c}$ , ( $b^2 - 4ac < 0$ ), then the minimum and maximum

values of  $f(x)$  are given by  $f\left(\pm\sqrt{\frac{c}{a}}\right)$

**W.E-20:** The maximum or minimum values of  $2x + 5 - 3x^2$  on  $R$  is

**Sol:** Given expression  $2x + 5 - 3x^2$ ,  
 $a = -3 < 0, b = 2, c = 5$   
 $\Rightarrow$  given expression has maximum value

$$\therefore \text{maximum value} = \frac{4ac - b^2}{4a} =$$

$$\frac{4(-3)(5) - 4}{4(-3)} = \frac{16}{3}$$

**W.E-21:** Find the range the expression  $\frac{x^2 - x + 1}{x^2 + x + 1}$ , where  $x \in R$ .

**Sol:** Here  $a=1, b=1, c=1$ , minimum value  $= f(1) = \frac{1}{3}$   
 maximum value  $= f(-1) = 3$

➤ If  $x$  is real then the maximum and minimum values of  $\frac{(a+x)(b+x)}{c+x}$  ( $x > -c, a > c, b > c$ ) are

$$\left(\sqrt{a-c} - \sqrt{b-c}\right)^2 \text{ and } \left(\sqrt{a-c} + \sqrt{b-c}\right)^2$$

➤ **Locating the roots of Quadratic Equation under given conditions:**

Location of real roots of the equation

$$f(x) = ax^2 + bx + c = 0 \quad (\Delta > 0) \text{ whose roots}$$

are  $\alpha, \beta$  ( $\alpha < \beta$ ) with respect to one real number  $K$ .

- If both roots are greater than  $k$  then  $\alpha + \beta > 2k$  and  $a f(k) > 0$
  - If both roots are less than  $k$ , then  $\alpha + \beta < 2k$  and  $a f(k) > 0$
  - If  $k$  lies between the roots then  $a f(k) < 0$
- with respect to two real numbers  $k_1, k_2$ .**

- If  $k_1 < \alpha < \beta < k_2$  then
  - i)  $a f(k_1) > 0$ , ii)  $a f(k_2) > 0$ , iii)  $2k_1 < \alpha + \beta < 2k_2$

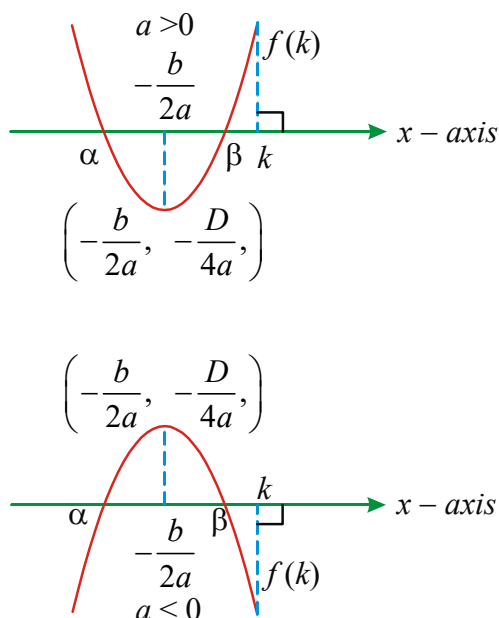
- If  $\alpha < k_1 < k_2 < \beta$  then
  - i)  $a f(k_1) < 0$ , ii)  $a f(k_2) < 0$

- If  $k_1 < \alpha < k_2 < \beta$  then
  - i)  $a f(k_1) > 0$  and ii)  $a f(k_2) < 0$

- **Position of roots of a quadratic equation:** Let  $f(x) = ax^2 + bx + c$ ,  $a \neq 0, b, c \in \mathbb{R}$ , and  $\alpha, \beta$  be the roots of  $f(x) = 0$ . Suppose  $k, k_1, k_2 \in \mathbb{R}$  and  $k_1 < k_2$ . Then, the following holds good

**(i) Conditions for a number  $k$  :**

[ If both the roots are less than  $k$  ]



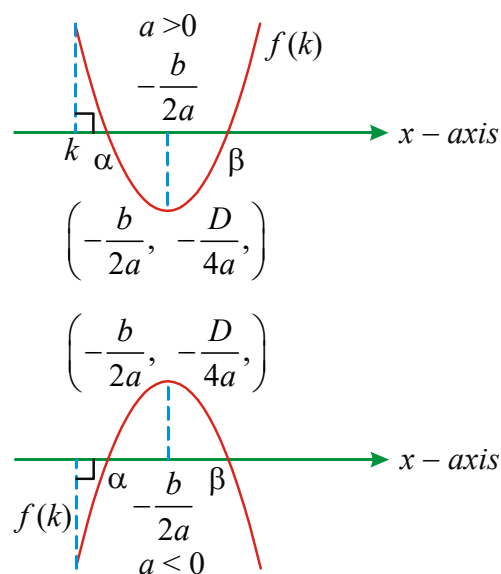
(a)  $D \geq 0$  ( roots may be equal )

(b)  $a f(k) > 0$

(c)  $k > -\frac{b}{2a}$ , where  $\alpha \leq \beta$

**(ii) Conditions for a number  $k$**

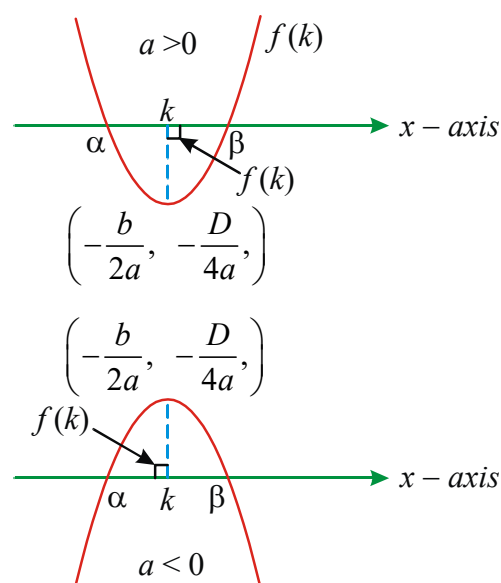
[If both the roots of  $f(x) = 0$  are greater than  $k$  ]



(a)  $D \geq 0$  (b)  $a f(k) > 0$  (c)  $k < -\frac{b}{2a}$

**(iii) Conditions for a number  $k$**

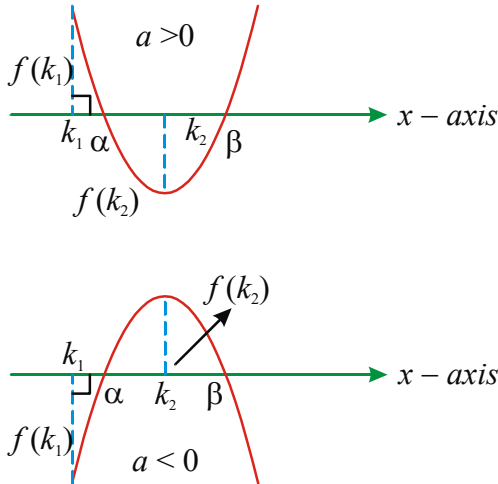
[ If  $k$  lies between the roots of  $f(x) = 0$  ]



(a)  $D > 0$  (b)  $a f(k) < 0$  where,  $\alpha < \beta$

(iv) Conditions for numbers  $k_1$  and  $k_2$  :

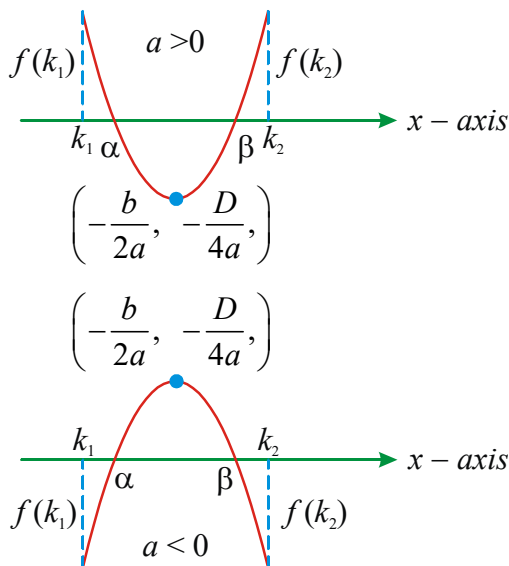
[ If exactly one root of  $f(x) = 0$  lies in the interval  $(k_1, k_2)$  ]



(a)  $D > 0$  (b)  $f(k_1)f(k_2) < 0$ , where  $\alpha < \beta$

(v) Conditions for numbers  $k_1$  and  $k_2$  :

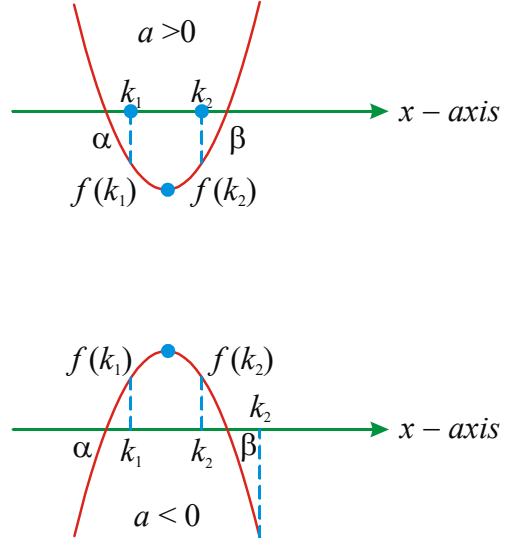
[ If both roots of  $f(x) = 0$  lies between  $k_1$  and  $k_2$  ]



- (a)  $D \geq 0$  ( roots may be equal )  
 (b)  $af(k_1) > 0$  (c)  $af(k_2) > 0$   
 (d)  $k_1 < -\frac{b}{2a} < k_2$  where  $\alpha \leq \beta$  and  $k_1 < k_2$ .

(vi) Conditions for numbers  $k_1$  and  $k_2$  :

[ If  $k_1$  and  $k_2$  lies between the roots of  $f(x) = 0$  ]

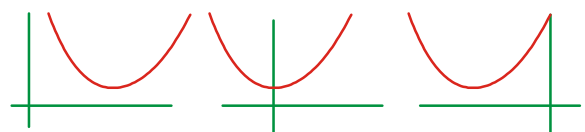


- (a)  $D > 0$  (b)  $af(k_1) < 0$   
 (c)  $af(k_2) < 0$ ,

**W.E-22:** If both roots of the quadratic equation  $x^2 - 2kx + k^2 + K - 5 = 0$  are less than 5, then find the interval in which  $k$  lies

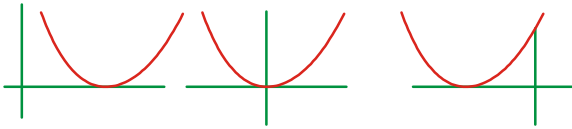
**Sol:**  $\Delta \geq 0 \Rightarrow 4k^2 - 4(k^2 + k - 5) \geq 0 \Rightarrow k \leq 5 \rightarrow (1)$   
 $af(k) \geq 0 \Rightarrow 1 \cdot f(5) > 0$   
 $\Rightarrow k^2 - 9k + 20 > 0 \Rightarrow k < 4$  or  $k > 5 \rightarrow (2)$   
 $k > \frac{-b}{2a} \Rightarrow 5 > \frac{-(-2k)}{2} \Rightarrow k < 5 \rightarrow (3)$   
 from (1), (2) and (3)  
 $k \in (-\infty, 4)$

➤ **Graphs of**  $y = ax^2 + bx + c$ ,  $ax^2 + bx + c = 0$   
 If  $a > 0$  and  $b^2 - 4ac < 0$  then the graph of  $y = ax^2 + bx + c$  lies entirely above the x-axis.

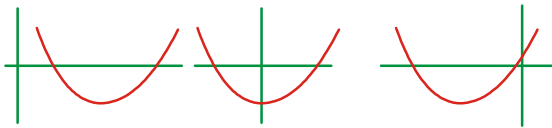




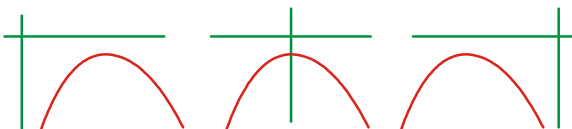
- If  $a > 0$  and  $b^2 - 4ac = 0$  then the graph of  $y = ax^2 + bx + c$  touches the x-axis and lies entirely above the x-axis.



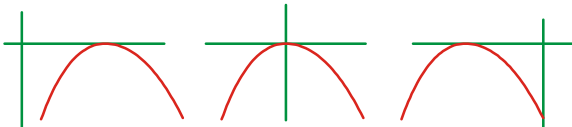
- If  $a > 0$  and  $b^2 - 4ac > 0$  then the graph of  $y = ax^2 + bx + c$  is an upward parabola which cuts the x-axis at two distinct points.



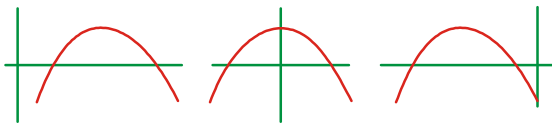
- If  $a < 0$  and  $b^2 - 4ac < 0$  then the graph of  $y = ax^2 + bx + c$  lies entirely below the x-axis.



- If  $a < 0$  and  $b^2 - 4ac = 0$  then the graph of  $y = ax^2 + bx + c$  lies entirely below the x-axis and touches the x-axis.



- If  $a < 0$  and  $b^2 - 4ac > 0$  then the graph of  $y = ax^2 + bx + c$  is a downward parabola which cuts the x-axis at two distinct points.



**W.E-23:** The interval in which 'a' lies when the graph of the function

$y = 16x^2 + 8(a+5)x - 7a - 5$  is strictly above the x-axis is

**Sol:**  $\Delta < 0 \Rightarrow 64(a+5)^2 + 64(7a+5) < 0$   
 $\Rightarrow (a+15)(a+2) < 0 \therefore -15 < a < -2$

- If the expression  $ax^2 + 2hxy + by^2 + 2gx + 2fy + c$  can be resolved into two linear factors then

$$\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$$

- **Different forms of Quadratic Equations (Direct and Indirect):**

- (i) If  $a^{f(x)} + b^{f(x)} = a + b$  and  $ab = 1$  then  $f(x) = \pm 1$

**W.E-24 :** If  $(7+4\sqrt{3})^{x^2-8} + (7-4\sqrt{3})^{x^2-8} = 14$  then the value of  $x$  is

**Sol:** Here  $(7+4\sqrt{3}) + (7-4\sqrt{3}) = 14$

$$\therefore x^2 - 8 = \pm 1; \quad x^2 - 8 = -1, x^2 - 8 = 1$$

$$x^2 = 7, \quad x^2 = 9; \quad x = \pm\sqrt{7}, x = \pm 3$$

- (ii) If  $a(f(x))^2 + bf(x) + c = 0$  Put  $f(x) = t$  which reduces to  $at^2 + bt + c = 0$  Now solve for  $t$
- (iii)  $a^{f(x)} = a^n$  then solve  $f(x) = n$
- (iv)  $a^{f(x)} + b^{f(x)} = a^n + b^n$  then solve  $f(x) = n$
- (v)  $(x-a)(x-b)(x-c)(x-d) = Ax^2$  and  $ab = cd$  then put  $t = x + \frac{ab}{x}$
- (vi)  $(x-a)(x-b)(x-c)(x-d) = A$  and  $a < b < c < d$ . Also  $b-a = d-c$  then  $t = x - \frac{(a+b+c+d)}{4}$
- (vii)  $(x \pm a)^4 + (x \pm b)^4 = P$  then put

$$t = \frac{(x \pm a) + (x \pm b)}{2} \text{ i.e. } t = x \pm \frac{(a+b)}{2}$$

$$\text{General } (\sqrt{x+m} + n)^4 + (\sqrt{x+m} - n)^4 = P$$

$$\text{then } x = -m \pm \sqrt{\frac{-6 \pm \sqrt{36 - 4n^4 + 2P}}{2}}$$



➤ **Facts to be remembered :**

- (i) For three consecutive integers  $a, b, c$  the equation  $f(x) = (x-a)^2 + (x-b)^2 + (x-c)^2 = 0$  has no real roots. Both roots should be imaginary and conjugate of each other.
- (ii) Any cubic equation of the form  $(x-\alpha)^3 + (x-\beta)^3 + (x-\gamma)^3 = 0$ ;  $\alpha, \beta, \gamma \in R$ , always possess one real and two imaginary roots as  $f'(x) > 0 \forall x \in R$ .
- (iii) Let  $ax^2 + bx + c = 0 \forall (a \neq 0, a, b, c \in R)$ . If roots of the equation are real and distinct then one root is more and one root is less than average of their sum i.e. if  $\alpha, \beta$  are real roots ( $\alpha \neq \beta$ ) of the equation  $ax^2 + bx + c = 0$  then  $\alpha < -\frac{b}{2a}, \beta > -\frac{b}{2a}$
- (iv)  $a, b, c > 0$ , then equation  $ax^2 + b|x| + c = 0$  has no real roots, in this case both roots will be imaginary and conjugate of each other.

**W.E-25:** The roots of  $x^2 + 5|x| + 4 = 0$  are

**Sol:** Write  $x^2 + 5|x| + 4 = 0$  as  $|x|^2 + 5|x| + 4 = 0$

$$\Rightarrow (|x|+1)(|x|+4) = 0$$

$$\Rightarrow |x| = -1, |x| = -4$$

(which is not possible if  $x \in R$ )

$\therefore$  the given equation has no real roots.

- (v) Number of quadratic equations which are unchanged by squaring their roots, is 4. They are quadratic equations whose roots are  $\{0, 0\}, \{0, 1\}, \{1, -1\}, \{\omega, \omega^2\}$  where  $\omega$  is a imaginary cube root of unity.
- (vi) If  $f(x) = ax^2 + bx + c = 0$  have no real roots and  $a + b + c < 0$  then  $c < 0$ .  
( $\because f(1) < 0 \Rightarrow a + b + c < 0 \Rightarrow f(0) < 0$ )



**CUQ**

- If  $a, b, c$  are positive then both roots of the equation  $ax^2 + bx + c = 0$** 
  - are real and negative
  - are real and positive
  - have negative real parts
  - have positive real parts
- In a quadratic equation  $ax^2 + bx + c = 0$  if 'a' and 'c' are of opposite signs and 'b' is real, then roots of the equation are**
  - real and distinct
  - real and equal
  - imaginary
  - both roots positive
- If one root of a quadratic equation is real and the other is imaginary, then the coefficients of the equation are**
  - real numbers
  - rational numbers
  - irrational numbers
  - complex numbers
- If  $a \neq b$  the roots of the equation  $(x-a)(x-b) = b^2$  are**
  - real and distinct
  - real and equal
  - real
  - imaginary
- If  $a, b, c$  are positive numbers in G.P. then the roots of the equation  $ax^2 + bx + c = 0$** 
  - are real and negative
  - have negative real parts
  - are equal
  - have negative imaginary parts
- If the ratio of the roots of the equation  $ax^2 + bx + c = 0$  is  $m : n$  then**
  - $\frac{m}{n} + \frac{n}{m} = \frac{b^2}{ac}$
  - $\sqrt{\left(\frac{m}{n}\right)} + \sqrt{\left(\frac{n}{m}\right)} = \frac{|b|}{\sqrt{ac}}$
  - $\sqrt{\left(\frac{m}{n}\right)} - \sqrt{\left(\frac{n}{m}\right)} = \frac{b^2}{ac}$
  - $\frac{m}{n} + \frac{n}{m} = \frac{a^2}{b^2}$
- If  $\alpha, \beta$  are the roots of the equation  $ax^2 + bx + c = 0$ , then the value of  $\frac{1}{a\alpha+b} + \frac{1}{a\beta+b} =$** 
  - $\frac{a}{bc}$
  - $\frac{b}{ac}$
  - $\frac{c}{ab}$
  - $\frac{ab}{c}$

8. If the roots of  $ax^2 + bx + c = 0$  and  $px^2 + qx + r = 0$  differ by the same quantity,

then  $\frac{b^2 - 4ac}{q^2 - 4pr} =$

1)  $\left(\frac{p}{a}\right)^2$  2)  $\left(\frac{c}{p}\right)^2$  3)  $\left(\frac{a}{p}\right)^2$  4)  $\left(\frac{p}{c}\right)^2$

9. If one root of the equation  $ax^2 + bx + c = 0$  is equal to the  $n^{\text{th}}$  power of the other, then

$(ac^n)^{1/n+1} + (a^n c)^{1/n+1} + b =$

1) 0 2) 1 3) -1 4) 2

10. If  $ax^2 + 2bx + c = 0$  and  $px^2 + 2qx + r = 0$  have one and only one root in common and  $a, b, c, p, q, r$  being rational, then  $b^2 - ac$  and  $q^2 - pr$  are

1) both are perfect squares

2)  $b^2 - ac$  is a perfect square but  $q^2 - pr$  is not a perfect square

3)  $q^2 - pr$  is a perfect square but  $b^2 - ac$  is not a perfect square

4) both are not perfect squares

11. If both the roots of  $ax^2 + bx + c = 0$  are positive then

1)  $\Delta > 0, ab > 0, ac > 0$

2)  $\Delta < 0, ab < 0, ac < 0$

3)  $\Delta > 0, ab < 0, ac > 0$

4)  $\Delta > 0, ab > 0, bc > 0$

12. If both the roots of  $ax^2 + bx + c = 0$  are negative then

1)  $\Delta > 0, ab > 0, bc < 0$

2)  $\Delta > 0, a, b, c$ , have the same signs

3)  $\Delta < 0, ab > 0, ac < 0$

4)  $\Delta < 0, ab > 0, bc > 0$

13. If both the roots of  $ax^2 + bx + c = 0$  are negative and  $b < 0$  then

1)  $a < 0, c < 0$  2)  $a < 0, c > 0$

3)  $a > 0, c < 0$  4)  $a > 0, c > 0$

14. If  $a > 0$ , then the expression  $ax^2 + bx + c$  is positive for all values of 'x' provided.

1)  $b^2 - 4ac > 0$

2)  $b^2 - 4ac < 0$

3)  $b^2 - 4ac = 0$

4)  $b^2 - ac < 0$

15. For real x, the expression  $\frac{(x-a)(x-b)}{(x-c)}$  will assume real values provided

1)  $a > b > c$

2)  $a < b < c$

3)  $c > a > b$

4)  $a < c < b$

16. If  $\alpha, \beta$  are the roots of  $ax^2 + bx + c = 0$  and  $k \in R$  then the condition so that  $\alpha < k < \beta$  is

1)  $ac > 0$

2)  $ak^2 + bk + c > 0$

3)  $ac < 0$

4)  $a^2k^2 + abk + ac < 0$

17. If  $a > 0$  and  $b^2 - 4ac < 0$ , then the graph of  $y = ax^2 + bx + c$

1) lies entirely below the x-axis

2) lies entirely above the x-axis

3) cuts the x-axis

4) touches the x-axis and lies below it

18. If  $a > 0$  and  $b^2 - 4ac = 0$ , then the graph of  $y = ax^2 + bx + c$

1) lies entirely above the x-axis

2) touches the x-axis and lies above it

3) touches the x-axis and lies below it

4) cuts the x-axis

19. If  $a < 0$  and  $b^2 - 4ac < 0$ , then the graph of  $y = ax^2 + bx + c$

1) lies entirely below the x-axis

2) lies entirely above the x-axis

3) cut the x-axis

4) touches the x-axis

20. If  $b^2 - 4ac > 0$  then the graph of  $y = ax^2 + bx + c$

1) cuts x-axis in two real points

2) touches the x-axis

3) lies entirely above the x-axis

4) can not be determined

21. If  $a, b, c$  are positive real numbers, then the number of real roots of the equation

$ax^2 + b|x| + c = 0$  is

1) 2

2) 4

3) 0

4) -1

22. Product of real roots of the equation

$t^2x^2 + |x| + 9 = 0$

1) is always positive 2) is always negative

3) does not exist

4)  $\frac{9}{t^2}$

23. If  $\alpha$  and  $\beta$ , ( $\alpha < \beta$ ), are the roots of the equation  $x^2 + bx + c = 0$ , where  $c < 0 < b$ , then

- 1)  $0 < \alpha < \beta$                       2)  $\alpha < 0 < \beta < |\alpha|$   
3)  $\alpha < \beta < 0$                       4)  $\alpha < 0 < |\alpha| < \beta$

24. If  $0 < a < b < c$ , and the roots  $\alpha, \beta$  of the equation  $ax^2 + bx + c = 0$  are non real complex roots, then

- 1)  $|\alpha| = |\beta|$                       2)  $|\alpha| > 1$   
3)  $|\beta| < 1$                       4)  $|\alpha| > 1, |\beta| < 1$

25. If  $b^2 \geq 4ac$  for the equation  $ax^4 + bx^2 + c = 0$ , then all roots of the equation will be real, if

- 1)  $b > 0, a > 0, c > 0$       2)  $b < 0, a > 0, c > 0$   
3)  $b > 0, a > 0, c < 0$       4)  $b < 0, a < 0, c < 0$

CUQ KEY

- 1) 3    2) 1    3) 4    4) 1    5) 2    6) 2  
7) 2    8) 3    9) 1    10) 1    11) 3    12) 2  
13) 1    14) 2    15) 4    16) 4    17) 2    18) 2  
19) 1    20) 1    21) 3    22) 3    23) 2    24) 1    25) 2

CUQ HINTS

- $\alpha + \beta = -\frac{b}{a} < 0, \alpha\beta = \frac{c}{a} > 0$
- $ac < 0 \Rightarrow \Delta = b^2 - 4ac > 0$
- If 1 root is real and other is complex number then the coefficients will be complex numbers.
- $\Delta = (a-b)^2 + 4b^2 > 0$
- $b^2 = ac \therefore b^2 - 4ac < 0 \therefore$  roots are imaginary  
. a, b, c are +ve  $\Rightarrow$  roots have -ve real parts.
- $\frac{b^2}{ac} = \frac{(m+n)^2}{mn} \Rightarrow \frac{m+n}{\sqrt{mn}} = \sqrt{\frac{b^2}{ac}}$
- $a\alpha + b = -c/\alpha; a\beta + b = -c/\beta$
- $|\alpha - \beta| = |\alpha_1 - \beta_1|$   
 $\sqrt{\frac{\Delta}{a^2}} = \sqrt{\frac{\Delta}{p^2}} \Rightarrow \frac{b^2 - 4ac}{q^2 - 4pr} = \frac{a^2}{p^2}$

9. roots are  $\alpha, \alpha^n$ ;  $\alpha + \alpha^n = \frac{-b}{a}, \alpha^{n+1} = \frac{c}{a}$ .

10.  $ax^2 + 2bx + c = 0$  and  $px^2 + 2qx + r = 0$  have one root common then it is rational other is also rational then  $b^2 - ac =$  perfect square and  $q^2 - pr =$  perfect square

11.  $\Delta > 0, ab < 0$  and  $ac > 0$   
12.  $\Delta > 0$  and  $ab > 0, ac > 0$   
13. a, b and c will have same sign  
14. roots are non real complex roots.

15. put  $y = \frac{(x-a)(x-b)}{(x-c)}$

$$x^2 - (a+b+y)x + ab + cy = 0$$

$$x \in R \Rightarrow \Delta = (a+b+y)^2 - 4(ab+cy) > 0$$

$$\Rightarrow y^2 + 2(a+b-2c)y + (a-b)^2 > 0$$

true for all  $y \in R$

$$\Rightarrow \Delta = 4(a+b-2c)^2 - 4(a-b)^2 < 0$$

$$\Rightarrow (c-a)(c-b) < 0$$

16.  $f(x) = ax^2 + bx + c$

$$a f(k) < 0 \Rightarrow a^2 k^2 + abk + ac < 0.$$

17.  $b^2 - 4ac < 0$  and  $a > 0$  then the graph lies entirely above the x-axis.

18.  $b^2 - 4ac = 0$  and  $a > 0$  then the graph of y touches the x-axis and lies above x-axis.

19.  $a < 0$  and  $b^2 - 4ac < 0$  the roots are imaginary then the graph lies entirely below the x-axis.

20.  $b^2 - 4ac > 0 \Rightarrow$  roots are real and distinct.  
 $\therefore$  graph cuts the x-axis in two distinct points.

21.  $ax^2 + b|x| + c > 0, \forall x$

22. real roots does not exist.

23.  $\alpha < \beta, c < 0 < b$ ;  $\alpha + \beta = -b, \alpha\beta = C$

C is negative and b is positive

$\alpha + \beta$  and  $\alpha\beta$  are negative

$\therefore$  one root is positive and other root is negative

$$\alpha < 0 < \beta \quad \alpha + \beta = -b \quad \beta = -b - \alpha \quad \beta < |\alpha|$$

24.  $0 < a < b < c$ ,  $\alpha + \beta = -\frac{b}{a}$  and  $\alpha\beta = \frac{c}{a}$

For non real complex roots  $b^2 - 4ac < 0$

$$\Rightarrow \frac{b^2}{a^2} - \frac{4c}{a} < 0$$

$$\Rightarrow (\alpha + \beta)^2 - 4\alpha\beta < 0$$

$$\Rightarrow (\alpha - \beta)^2 < 0$$

$$\therefore 0 < a < b < c$$

$\therefore$  roots are conjugate, then  $|\alpha| = |\beta|$

25. All roots of equation  $ax^4 + bx^2 + c = 0$  will be real if both roots of  $ay^2 + by + c = 0$  will be +ve

(let  $x^2 = y$ ), ie, sum of roots  $= -\frac{b}{a} > 0$  and

product of roots  $= \frac{c}{a} > 0$ .

Hence, a and b are of opposite sign, while a and c are of same sign.



### LEVEL-I (C.W)

### RELATION BETWEEN ROOTS AND COEFFICIENTS

1. If  $\alpha, \beta$  are roots of  $ax^2 + bx + c = 0$  then

$$\frac{1}{\alpha^3} + \frac{1}{\beta^3} =$$

1)  $\frac{3abc - b^3}{a^3}$       2)  $\frac{3ab - b^3}{a^2c}$

3)  $\frac{3abc - b^3}{c^3}$       4)  $\frac{b^2 - 2ac}{ac}$

2. If  $\alpha, \beta$  are roots of the equation

$2x^2 + 6x + b = 0 (b < 0)$  then  $\frac{\alpha}{\beta} + \frac{\beta}{\alpha}$  is less than

1) 2      2) -2      3) 18      4) 0

3. If  $\alpha, \beta$  are the roots of  $x^2 - p(x+1) + C = 0$  then  $(\alpha+1)(\beta+1) =$

1) 1-C      2) 1+C      3) C-1      4) C

### ROOTS AND/OR CONDITIONS GIVEN

4. If  $\omega$  is an imaginary cube root of unity. Then the equation whose roots are  $2\omega + 3\omega^2, 2\omega^2 + 3\omega$  is

1)  $x^2 + 5x + 7 = 0$       2)  $x^2 + 5x - 7 = 0$

3)  $x^2 - 5x + 7 = 0$       4)  $x^2 - 5x - 7 = 0$

5. If p and q are the roots of  $x^2 + px + q = 0$  then

1)  $p = 1$       2)  $p = 1$  or 0

3)  $p = -2$       4)  $p = -2$  or 0

### NATURE OF THE ROOTS AND PROPERTIES

6. If one root of  $x^2 - (3+2i)x + (1+3i) = 0$  is  $1+i$  then the other root is

1)  $1-i$       2)  $2+i$       3)  $3+i$       4)  $1+3i$

7. If the expression

$x^2 - (5m-2)x + (4m^2 + 10m + 25)$  can be expressed as a perfect square, then m =

1)  $\frac{8}{3}$  or 4      2)  $-\frac{8}{3}$  or 4

3)  $\frac{4}{3}$  or 8      4)  $-\frac{4}{3}$  or 8

8. If the roots of the quadratic equation  $x^2 - 4x - \log_3 a = 0$  are real, then the least value of a is

1) 81      2)  $\frac{1}{81}$       3)  $\frac{1}{64}$       4) 9

9. Roots of the equation  $2x^2 - 5x + 1 = 0$  and  $x^2 + 5x + 2 = 0$  are

1) Reciprocal and of the same sign

2) Reciprocal and of opposite sign

3) Equal in magnitude      4) Imaginary

### SOLVING EQUATIONS

10. If  $3^{1+x} + 3^{1-x} = 10$  then the values of x are

1) 1, -1      2) 1, 0      3) 1, 2      4) -1, -2

11. If  $(x+1)$  is a factor of

$$x^4 + (p-3)x^3 - (3p-5)x^2 + (2p-9)x + 6$$

then the value of p is

1) -4      2) 0      3) 4      4) 2

12. The number of real solutions of the equation  $\sin(e^x) = 5^x + 5^{-x}$  is (are)  
1) 0      2) 1      3) 2      4) Infinitely many
13. The roots of the equation  $x^{\sqrt{x}} = (\sqrt{x})^x$  are  
1) 0 and 1      2) 0 and 4  
3) 1 and 4      4) 0, 1 and 4

COMMON ROOTS

14. If  $(x-2)$  is a common factor of the expressions  $x^2 + ax + b$  and  $x^2 + cx + d$ , then  $\frac{b-d}{c-a} =$   
1) -2      2) -1      3) 1      4) 2

SIGN OF THE EXPRESSIONS AND INEQUALITIES

15. If  $x < 5$ , then the sign of the expression  $2x + 7 - 5x^2$  is  
1) positive      2) negative  
3) non negative      4) can not say
16. If  $x$  is real and  $5x^2 + 2x + 9 > 3x^2 + 10x + 7$ , then  $x$  lies in the interval  
1)  $(2 - \sqrt{3}, 2 + \sqrt{3})$   
2)  $(-\infty, 2 - \sqrt{3}) \cup (2 + \sqrt{3}, \infty)$   
3)  $(\sqrt{2} - 1, \sqrt{2} + 1)$   
4)  $(2 + \sqrt{3}, \infty)$

MAXIMUM AND MINIMUM

17. Maximum value of  $-5x^2 + 2x + 3$  is  
1)  $\frac{14}{5}$       2)  $\frac{13}{5}$       3)  $\frac{12}{5}$       4)  $\frac{16}{5}$
18. The maximum value of  $\frac{1}{4x^2 + 2x + 1}$  is  
1)  $\frac{4}{3}$       2)  $\frac{2}{3}$       3) 1      4)  $\frac{3}{4}$
19. If  $x \in [2, 4]$  then for the expression  $x^2 - 6x + 5$   
1) the least value = -4      2) the greatest value = 4  
3) the least value = 3  
4) the greatest value = -5

MODULUS FUNCTIONS

20. Number of solutions of the equation  $|x|^2 - 3|x| + 2 = 0$  is.....  
1) 4      2) 2      3) 0      4) 1
21. For the equation  $|x|^2 + |x| - 6 = 0$  the roots are  
1) one and only one real number  
2) real with sum one  
3) real with sum zero  
4) real with product zero

TRANSFORMED EQUATIONS

22. The equation formed by increasing each root of  $ax^2 + bx + c = 0$  by 1 is  $2x^2 + 8x + 2 = 0$  then  
1)  $a+b=0$       2)  $b+c=0$       3)  $b=c$       4)  $a=b$

QUADRATIC EXPRESSION IN TWO VARIABLES

23. The condition for  $ax^2 + 2cxy + by^2 + 2bx + 2ay + c$  is resolvable into two linear factors is  
1)  $a^3 + b^3 + c^3 = 3abc$       2)  $a^3 + b^3 + c^3 = abc$   
3)  $a^3 + b^3 + c^3 = ab + bc + ca$       4)  $a^3 + b^3 + c^3 + abc = 0$
24. If  $x^2 + 4xy + 4y^2 + 4x + cy + 3$  can be written as the product of two linear factors, then  $c =$   
1) 2      2) 4      3) 6      4) 8

LEVEL-I (C.W.) - KEY

- |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|
| 1) 3  | 2) 2  | 3) 2  | 4) 1  | 5) 2  | 6) 2  |
| 7) 4  | 8) 2  | 9) 2  | 10) 1 | 11) 3 | 12) 1 |
| 13) 3 | 14) 4 | 15) 4 | 16) 2 | 17) 4 | 18) 1 |
| 19) 1 | 20) 1 | 21) 3 | 22) 3 | 23) 1 | 24) 4 |

LEVEL-I (C.W.) - HINTS

1.  $\frac{1}{\alpha^3} + \frac{1}{\beta^3} = \frac{(\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)}{(\alpha\beta)^3}$
2.  $\alpha + \beta = -3, \alpha\beta = b/2$   
 $D = 36 - 8b > 0 (\because b < 0)$   
 $\frac{\alpha}{\beta} + \frac{\beta}{\alpha} = \frac{\alpha^2 + \beta^2}{\alpha\beta} = \frac{(\alpha + \beta)^2 - 2\alpha\beta}{\alpha\beta}$   
 $\frac{(\alpha + \beta)^2}{\alpha\beta} - 2 = \frac{18}{b} - 2 < -2$

3.  $\alpha\beta + \alpha + \beta + 1 = -p + c + p + 1$
4.  $\alpha + \beta = 2(\omega + \omega^2) + 3(\omega^2 + \omega) = -5$   
 $\alpha\beta = (2\omega + 3\omega^2)(2\omega^2 + 3\omega) = 7$
5.  $p + q = -p, pq = q$   
 $\Rightarrow q = 0, p = 1 \Rightarrow p = 0, q = -2$
6.  $\alpha + \beta = 3 + 2i$
7.  $\Delta = 0$
8.  $D \geq 0 \Rightarrow \log_3 a \geq -4 \Rightarrow a \geq 3^{-4} \Rightarrow a \geq \frac{1}{81}$
9.  $f(x) = 2x^2 - 5x + 1 = 0$   
 $f\left(-\frac{1}{x}\right) = 0 \Rightarrow x^2 + 5x + 2 = 0$
10. Put  $3^x = t$
11. Since  $x+1$  is a factor of  $f(x)$ ,  
 $-1$  must be a root of  $f(x) = 0$   
 $\therefore f(-1) = 0 \Rightarrow -6p + 24 = 0 \Rightarrow p = 4$
12.  $5^x + 5^{-x} \geq 2$  but  $\sin(e^x) \leq 1$
14.  $x - 2$  is a common factor  $\Rightarrow x = 2$  is common root  
 $\therefore 4 + 2a + b = 0$  and  $4 + 2c + d = 0$   
 $\Rightarrow 2(a - c) + (b - d) = 0 \Rightarrow \frac{b - d}{c - a} = 2$
15.  $5x^2 - 2x - 7 = 0$  ;  
 $5x^2 + 5x - 7x - 7 = 0 \quad (x+1)(5x-7) = 0$
16.  $2x^2 - 8x + 2 > 0$  ;  $x^2 - 4x + 1 > 0$   
 $\text{root} = \frac{4 \pm \sqrt{16 - 4}}{2} = 2 \pm \sqrt{3}$   
 $= (-\infty, 2 - \sqrt{3}) \cup (2 + \sqrt{3}, \infty)$
17.  $\frac{4ac - b^2}{4a}$
18. Maximum value of  $\left(\frac{1}{4x^2 + 2x + 1}\right)$   
 $= \frac{1}{\min \text{ of } (4x^2 + 2x + 1)} = \frac{1}{\frac{4ac - b^2}{4a}}$
19.  $x^2 - 6x + 5 = (x - 3)^2 - 4 \geq -4$

20.  $(|x| - 2)(|x| - 1) = 0 \Rightarrow x = \pm 1, \pm 2$
21.  $(|x| + 3)(|x| - 2) = 0 \Rightarrow x = \pm 2$
22.  $(\alpha + 1)(\beta + 1) = 1 \Rightarrow \alpha\beta = -(\alpha + \beta)$   
 $\Rightarrow \frac{c}{a} = \frac{b}{a} \Rightarrow b = c$
23.  $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$
24.  $\Delta \equiv abc + 2fgh - af^2 - bg^2 - ch^2 = 0$



LEVEL-I (H.W)

RELATION BETWEEN ROOTS AND COEFFICIENTS

1. If  $\alpha, \beta$  are the roots of  $x^2 + 3x - 2 = 0$  then  
 $\frac{\alpha + 1}{\beta} + \frac{\beta + 1}{\alpha} =$   
 1)  $\frac{-15}{2}$     2)  $\frac{6}{5}$     3)  $\frac{-5}{2}$     4)  $-5$
2. If  $\alpha, \beta$  are the roots of  $x^2 + bx + c = 0$  and  
 $\alpha + h, \beta + h$  are the roots of  $x^2 + qx + r = 0$   
 then  $h =$   
 1)  $b + q$     2)  $b - q$     3)  $\frac{1}{2}(b + q)$     4)  $\frac{1}{2}(b - q)$
3. If the sum of the roots of the equation  
 $5x^2 - 4x + 2 + k(4x^2 - 2x - 1) = 0$  is 6, then  $k =$   
 1)  $13/17$     2)  $17/13$     3)  $-17/13$     4)  $-13/11$
4. If  $\alpha, \beta$  are the roots of the equation  
 $x^2 + 3x + 2 = 0$ . Then  $\alpha^5 + \beta^5 =$   
 1) 33    2)  $-32$     3) 35    4)  $-33$
5. The quadratic equation whose roots are  
 $\frac{\sqrt{2} + \sqrt{3}i}{\sqrt{2} - \sqrt{3}i}$  and  $\frac{\sqrt{2} - \sqrt{3}i}{\sqrt{2} + \sqrt{3}i}$   
 1)  $5x^2 - 2x + 5 = 0$     2)  $5x^2 + 2x + 5 = 0$   
 3)  $5x^2 + 2x - 5 = 0$     4)  $5x^2 - 2x - 5 = 0$
6. Let two numbers have arithmetic mean 9 and  
 geometric mean 4. Then these numbers are  
 the roots of the quadratic equation  
 1)  $x^2 + 18x + 16 = 0$     2)  $x^2 - 18x - 16 = 0$   
 3)  $x^2 + 18x - 16 = 0$     4)  $x^2 - 18x + 16 = 0$

7. If the ratio of the roots of  $x^2 + bx + c = 0$  and  $x^2 + qx + r = 0$  is same then

- 1)  $r^2c = qb^2$       2)  $r^2b = qc^2$   
3)  $rb^2 = cq^2$       4)  $rc^2 = bq^2$

### NATURE OF THE ROOTS AND PROPERTIES

8. If the roots of the equation  $(a^2 + b^2)x^2 + 2(bc + ad)x + (c^2 + d^2) = 0$  are real then  $a^2, bd, c^2$  are in  
1) A.P    2) G.P    3) H.P    4) A.G.P
9. The roots of the equation  $x^2 - 2\sqrt{3}x + 3 = 0$  are  
1) rational and equal    2) rational and not equal  
3) irrational    4) imaginary
10. If '1' is one of the roots of  $ax^2 + 3x + 5 = 0$  then the second root is  
1)  $\frac{5}{3}$     2)  $-\frac{5}{3}$     3)  $\frac{5}{16}$     4)  $-\frac{5}{8}$
11. If '3' is root of  $x^2 + kx - 24 = 0$  then it is also root of  
1)  $x^2 + 5x + k = 0$     2)  $x^2 + kx + 24 = 0$   
3)  $x^2 - kx + 6 = 0$     4)  $x^2 - 5x + k = 0$
12. If  $4 - i\sqrt{3}$  is a root of quadratic equation then the equation is  
1)  $x^2 - 8x + 13 = 0$     2)  $x^2 - 8x + 19 = 0$   
3)  $x^2 - 8x - 13 = 0$     4)  $x^2 - 8x - 19 = 0$
13. If the product of the roots of the equation  $5x^2 - 4x + 2 + k(4x^2 - 2x - 1) = 0$  is 2 then  $k =$   
1)  $-\frac{8}{9}$     2)  $\frac{8}{9}$     3)  $\frac{4}{9}$     4)  $-\frac{4}{9}$

### COMMON ROOTS

14. If the equations  $x^2 + ax + b = 0$  and  $x^2 + bx + a = 0 (a \neq b)$  have a common root then  $a + b =$   
1) -1    2) 0    3) 1    4) 2

### MAXIMUM AND MINIMUM

15. If  $x$  is real then the maximum value of the expression  $5 + 4x - 4x^2$  is  
1) 6    2) -6    3) 12    4) -12
16. The minimum value of the quadratic expression  $x^2 + bx + c$  is  
1)  $cb^2$     2)  $c^2b$     3)  $c + b^2$     4)  $c - \frac{b^2}{4}$
17. If  $2x - 7 - 5x^2$  has maximum value at  $x = a$  then  $a =$   
1)  $-\frac{1}{5}$     2)  $\frac{1}{5}$     3)  $\frac{34}{5}$     4)  $-\frac{34}{5}$

### MODULUS FUNCTIONS

18. If  $a, b, c$  are positive real numbers then the number of real roots of the equation  $ax^2 + b|x| + c = 0$  is  
1) 2    2) 4    3) 0    4) -1
19. Number of real roots of the equation  $3x^2 + 4|x| + 5 = 0$  are  
1) 2    2) 4    3) 0    4) -1

### TRANSFORMED EQUATIONS

20. The equation formed by decreasing each root of the equation  $ax^2 + bx + c = 0$  by 1 is  $2x^2 + 8x + 2 = 0$  then  
1)  $a = -b$     2)  $b = -c$     3)  $c = -a$     4)  $b = a + c$
21. If  $\alpha$  and  $\beta$  are the roots of  $x^2 - 2x + 4 = 0$  then the value of  $\alpha^6 + \beta^6$  is (EAM - 2009)  
1) 32    2) 64    3) 128    4) 256

### LEVEL-I (H.W.) - KEY

- 1) 4    2) 4    3) 4    4) 4    5) 2    6) 4  
7) 3    8) 2    9) 3    10) 4    11) 3    12) 2  
13) 1    14) 1    15) 1    16) 4    17) 2    18) 3  
19) 3    20) 2    21) 3

### LEVEL-I (H.W.) - HINTS

1.  $\alpha + \beta = -3, \alpha\beta = -2$
2.  $\alpha + h + \beta + h = -q \Rightarrow 2h = -q - (\alpha + \beta)$
3. Sum of the roots  $= 6 \Rightarrow \frac{2k+4}{5+4k} = 6 \Rightarrow k = \frac{-13}{11}$



4.  $\alpha = -2, \beta = -1$

5.  $\alpha + \beta = \frac{2(2-3)}{2+3} : \alpha\beta = 1$

The equation is  $5x^2 + 2x + 5 = 0$

6. Let the two numbers be  $\alpha$  and  $\beta$ . Given that

$$\sqrt{\alpha\beta} = 4, \alpha + \beta = 18, \alpha\beta = 16$$

$\therefore$  Required equation is  $x^2 - 18x + 16 = 0$

7.  $\frac{b_1^2}{a_1c_1} = \frac{b_2^2}{a_2c_2}$

8.  $(ax+d)^2 + (bx+c)^2 = 0$

$$ax+d=0 \text{ and } bx+c=0$$

$$\frac{-d}{a} = \frac{-c}{b} = x \Rightarrow ac = bd$$

$$\Rightarrow a^2c^2 = (bd)^2$$

9.  $\Delta = 12 - 12 = 0$  roots are  $-\sqrt{3}, -\sqrt{3}$

10.  $a+3+5=0$ , replace a value find roots

11.  $9+3k-24=0 \Rightarrow K=5$

12.  $4-i\sqrt{3}, 4+i\sqrt{3}$  are the roots  $\alpha + \beta = 8$  and  $\alpha\beta = 19$

13.  $\frac{2-k}{5+4k} = 2; 2-10=8k+k; k = \frac{-8}{9}$

14.  $\alpha^2 + a\alpha + b = 0$  and  $\alpha^2 + ba + a = 0$   
 $\Rightarrow \alpha(a-b) + (b-a) = 0 \Rightarrow \alpha = 1$

15.  $\frac{4ac-b^2}{4a} = \frac{-80-16}{4(-4)} = \frac{-96}{-16} = 6$

16. Use formula  $= \frac{4ac-b^2}{4a}$

17.  $x = -\frac{b}{2a} = -\frac{2}{2(-5)} = \frac{1}{5}$

18.  $\alpha^2 + b|x| + c > 0, \forall x \in R$

19. L.H.S is positive for all  $x \in R$

20.  $f(x+1) = 0$

21.  $\alpha = -2\omega$   
 $\beta = -2\omega$



## LEVEL-II (C.W)

### RELATION BETWEEN ROOTS AND COEFFICIENTS

1. If  $\alpha, \beta$  are the roots of  $ax^2 + bx + c = 0$  then

the value  $\left(\frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right)^2$  is

1)  $\frac{b^2(b^2-4ac)}{c^2a^2}$  2)  $\frac{b^2(b^2+4ac)}{c^2a^2}$

3)  $\frac{b^2(b^2-2ac)}{c^2a^2}$  4)  $\frac{b^2(b^2+2ac)}{c^2a^2}$

2. Let two numbers have arithmetic mean 9 and geometric mean 4. Then these numbers are the roots of the quadratic equation

1)  $x^2 + 18x + 16 = 0$  2)  $x^2 - 18x - 16 = 0$

3)  $x^2 + 18x - 16 = 0$  4)  $x^2 - 18x + 16 = 0$

3. If  $\alpha \neq \beta$  but  $\alpha^2 = 2\alpha - 3; \beta^2 = 2\beta - 3$  then

the equation whose roots are  $\frac{\alpha}{\beta}$  and  $\frac{\beta}{\alpha}$  is

1)  $2x^2 + 3x + 2 = 0$  2)  $3x^2 + 2x + 3 = 0$

3)  $2x^2 - 3x + 2 = 0$  4)  $3x^2 - 2x + 3 = 0$

4. If  $\alpha + \beta = -2$  and  $\alpha^3 + \beta^3 = -56$  then the quadratic equation whose roots are  $\alpha, \beta$  is  
 [EAM 2008]

1)  $x^2 + 2x - 16 = 0$  2)  $x^2 + 2x - 15 = 0$

3)  $x^2 + 2x - 12 = 0$  4)  $x^2 + 2x - 8 = 0$

5. If the sum of the roots of the equation  $ax^2 + bx + c = 0$  is equal to sum of the squares of their reciprocals, then  $bc^2, ca^2, ab^2$  are in

1) A.P 2) G.P 3) H.P 4) A.G.P

6.  $\tan 22^\circ$  and  $\tan 23^\circ$  are roots of  $x^2 + ax + b = 0$  then

1)  $a+b+1=0$  2)  $a-b+1=0$

3)  $b-a+1=0$  4)  $a+b=1$

7. If the difference of the squares of the roots of the equation  $x^2 - 6x + q = 0$  is 24, then the value of q is

1) -7 2) 8 3) 5 4) 4

8. Let  $\alpha, \beta$  be the roots of the equation  $(x-a)(x-b)=c, c \neq 0$ , then the roots of the equation  $(x-\alpha)(x-\beta)+c=0$  are  
 1)  $a, c$                                   2)  $b, c$   
 3)  $a, b$                                   4)  $a+b, b+c$
9. If  $\alpha, \beta$  are the roots of the equation  $x^2-15x+1=0$ , then the value of  $\left(\frac{1}{\alpha}-15\right)^{-2} + \left(\frac{1}{\beta}-15\right)^{-2}$  is  
 1) 225    2) 900    3) 223    4) 0

**ROOTS ARE GIVEN WITH CONDITIONS**

10. The quadratic equation whose roots are the x and y intercepts of the line passing through (1,1) and making a triangle of area A with the axes, may be  
 1)  $x^2 + Ax + 2A = 0$     2)  $x^2 - 2Ax + 2A = 0$   
 3)  $x^2 - Ax + 2A = 0$     4)  $x^2 - 2Ax - 2A = 0$
11. If one root of  $x^2 + px + 12 = 0$  is 4 while the equation  $x^2 + px + q = 0$  has equal roots then  $q =$   
 1) -7    2) 4    3) 42    4)  $\frac{49}{4}$
12. If  $k > 0$  and the product of the roots of the equation  $x^2 - 3kx + 2e^{2 \log k} - 1 = 0$  is 7 then the sum of the roots is  
 1) 2    2) 4    3) 6    4) 8
13. The value of 'a' for which one root of the quadratic equation  $(\alpha^2 - 5a + 3)x^2 - (3a - 1)x + 2 = 0$  is twice as large as other is  
 1) -2/3    2) 1/3    3) -1/3    4) 2/3
14. Two students while solving a quadratic equation in x, one copied the constant term incorrectly and got the roots as 3 and 2. The other copied the constant term and coefficient of  $x^2$  as -6 and 1 respectively. The correct roots are  
 1) 3, -2                                  2) -3, 2  
 3) -6, -1                                  4) 6, -1

15. Let A, G and H be the A.M, G.M and H.M of two positive numbers a and b. The quadratic equation whose roots are A and H is  
 1)  $Ax^2 - (A^2 + G^2)x + AG^2 = 0$   
 2)  $Ax^2 - (A^2 + H^2)x + AH^2 = 0$   
 3)  $Hx^2 + (H^2 + G^2)x + HG^2 = 0$   
 4)  $Gx^2 - (H^2 + G^2)x + GH^2 = 0$
16. If one root of the equation  $ax^2 + bx + c = 0$  is the square of the other, then  
 1)  $b^2 + ac^2 + a^2c = 3abc$   
 2)  $b^3 + ac^2 + a^2c = 3abc$   
 3)  $b^2 + ac^2 + a^2c + 3abc = 0$   
 4)  $b^3 + ac^2 + a^2c + 3abc = 0$
17. The roots of the equation  $ax^2 + bx + c = 0, a \in \mathbb{R}^+$ , are two consecutive odd positive integers. Then  
 1)  $|b| \leq 4a$     2)  $|b| \geq 4a$     3)  $|b| \geq 2a$     4)  $|b| \leq a$
18. The number of real solution of the equation  $\left(\frac{9}{10}\right)^x = -3 + x - x^2$  is  
 1) 2    2) 0    3) 1    4) 3
19. If the roots of  $x^2 - bx + c = 0$  are two consecutive integers then  $b^2 - 4c =$   
 1) 0    2) 1    3) 2    4) 3
20. If the roots of the quadratic equation  $x^2 + px + q = 0$  are  $\tan 30^\circ$  and  $\tan 15^\circ$  respectively then the value of  $2 + q - p$  is  
 1) 1    2) 2    3) 3    4) 0

**NATURE OF THE ROOTS AND PROPERTIES**

21. If  $a + b + c = 0$  then the equation  $3ax^2 + 2bx + c = 0$  has  
 1) imaginary roots                      2) real and equal roots  
 3) real and different roots    4) rational roots
22. If the roots of the equation  $(a^2 + b^2)x^2 + 2(bc + ad)x + (c^2 + d^2) = 0$  are real then  $a^2, bd, c^2$  are in  
 1) AP    2) GP    3) HP    4) AGP

23. Both the roots of the equation  $(x-b)(x-c) + (x-c)(x-a) + (x-a)(x-b) = 0$  are  
 1) positive 2) negative  
 3) real 4) imaginary
24. If  $a, b, c$  are in A.P. then the roots of the equation  $ax^2 + 2bx + c = 0$  are  
 1) real and distinct 2) real and equal  
 3) real 4) imaginary
25. If  $ax^2 - (2a+3)x + (3+5a) = 0$  has no real roots, then 'a' lies in the interval  
 1)  $\left(-\frac{3}{4}, \frac{3}{4}\right)$  2)  $\left(-\infty, -\frac{3}{4}\right)$   
 3)  $\left(-\infty, \frac{3}{4}\right)$  4)  $\left(-\infty, -\frac{3}{4}\right) \cup \left(\frac{3}{4}, \infty\right)$
26. If the roots of  $(x-a)(x-b) + (x-b)(x-c) + (x-c)(x-a) = 0$  are equal then which of the following is not possible  
 1)  $a+b+c=0$  2)  $a=b=c$   
 3)  $a+bw+cw^2=0$  4)  $a+bw^2+cw=0$
27. If the roots of  $ax^2 - bx - c = 0$  are changed by the same quantity then the expression in  $a, b, c$  that does not change is  
 1)  $\frac{b^2 - 4ac}{a^2}$  2)  $\frac{b^2 - 4ac}{a}$   
 3)  $\frac{b^2 + 4ac}{a^2}$  4)  $\frac{b^2 + 4ac}{a}$
28. If the roots of the equation  $bx^2 + cx + a = 0$  be imaginary then for all real values of  $x$ , the expression  $3b^2x^2 + 6bcx + 2c^2$  is  
 1) greater than  $4ab$  2) less than  $4ab$   
 3) greater than  $-4ab$  4) less than  $-4ab$

### SOLVING EQUATIONS

29. Let  $f(x) = x^2 - 3x + 4$ , the value(s) of  $x$  which satisfies  $f(1) + f(x) = f(1)f(x)$  is  
 1) 1 2) 2  
 3) 1 or 2 4) 1 and 0

30. The number of non-zero solutions of the equation  $x^2 - 5x - (\operatorname{sgn} x)6 = 0$  is  
 1) 1 2) 2 3) 3 4) 4
31. A root of equation  $\frac{a+c}{x+a} + \frac{b+c}{x+b} = \frac{2(a+b+c)}{x+a+b}$  is  
 1) a 2) b 3) c 4)  $a+b+c$
32. The equation  $\log_2(3-x) + \log_2(1-x) = 3$  has  
 1) One root 2) Two roots  
 3) Infinite roots 4) No root
33. The number of real roots of  $\sin(2^x)\cos(2^x) = \frac{1}{4}(2^x + 2^{-x})$  is  
 1) 1 2) 2 3) 3 4) No solution
34. The equation  $e^{\sin x} - e^{-\sin x} - 4 = 0$  has  
 1) infinite number of real roots  
 2) no real roots  
 3) exactly one real root  
 4) exactly four real roots
35. The real value of 'a' for which  $x^2 + i(a-1)x + 5 = 0$  will have a pair of conjugate complex roots is  
 1) values satisfying  $a^2 - 2a + 21 > 0$   
 2) 1 3) all values 4) no value

### COMMON ROOTS

36. If  $ax^2 + bx + c = 0$  and  $bx^2 + cx + a = 0$  have a common root and  $a \neq 0$  then  $\frac{a^3 + b^3 + c^3}{abc} =$   
 1) 1 2) 2 3) 3 4) 9
37. The equations  $x^2 + 3x + 5 = 0$  and  $ax^2 + bx + c = 0$  have a common root. If  $a, b, c \in \mathbb{N}$  then the least possible values of  $a+b+c$  is equal to  
 1) 3 2) 6 3) 9 4) 12
38. If the equations  $ax^2 + 2bx + 3c = 0$  and  $3x^2 + 8x + 15 = 0$  have a common root where  $a, b, c$  are the length of the sides of  $\triangle ABC$  then  $\sin^2 A + \sin^2 B + \sin^2 C$  is equal to  
 1) 1 2)  $\frac{3}{2}$  3)  $\sqrt{2}$  4) 2

LOCATION OF ROOTS

39. The set of values of  $p$  for which 6 lie between the roots of the equation  $x^2 + 2(p-3)x + 9 = 0$  is

1)  $\left(\frac{3}{4}, \infty\right)$  2)  $\left(-\infty, -\frac{3}{4}\right)$  3)  $\left(-\frac{3}{4}, \frac{3}{4}\right)$  4)  $\left[-\frac{3}{4}, \frac{3}{4}\right]$

40. All the values of  $m$  for which both roots of the equation  $x^2 - 2mx + m^2 - 1 = 0$  are greater than -2 but less than 4, lie in the interval

1)  $-1 < m < 3$  2)  $1 < m < 4$   
3)  $-2 < m < 0$  4)  $m > 3$

41. If  $a \in R$  and the roots of  $x^2 - 2x - a^2 + 1 = 0$  lies between the roots of

$x^2 - 2(a+1)x + a(a-1) = 0$  then 'a' belongs to

1)  $(1, \infty)$  2)  $\left(-\frac{1}{4}, 1\right)$  3)  $(-\infty, 0)$  4)  $\left(-\infty, -\frac{1}{4}\right)$

SIGN OF THE EXPRESSIONS AND INEQUALITIES

42. If the difference between the roots of the equation  $x^2 + ax + 1 = 0$  is less than  $\sqrt{5}$ , then the set of possible values of  $a$  is

1)  $(-3, 3)$  2)  $(-3, \infty)$  3)  $(3, \infty)$  4)  $(-\infty, -3)$

43. The greatest negative integer satisfying  $x^2 + 4x - 77 < 0$  and  $x^2 > 4$  is

1) -1 2) -2 3) -3 4) -10

44. The solution set contained in  $R$  of the inequality  $3^x + 3^{1-x} - 4 < 0$  is

1)  $(1, 3)$  2)  $(0, 1)$  3)  $(1, 2)$  4)  $(0, 2)$

45. If the equation  $ax^2 + 2bx - 3c = 0$  has non-real roots and  $\frac{3c}{4} < a + b$  then  $c$  is always

1)  $< 0$  2)  $> 0$  3)  $\geq 0$  4) zero

46. The equation  $(\cos p - 1)x^2 + (\cos p)x + (\sin p) = 0$  in the variable  $x$  has real roots. Then  $p$  can take any value in the interval

1)  $(0, 2\pi)$  2)  $(-\pi, 0)$  3)  $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$  4)  $(0, \pi)$

MAXIMUM AND MINIMUM OF QUADRATIC EXPRESSION

47. If  $a$  and  $b (\neq 0)$  are the roots of the equation  $x^2 + ax + b = 0$  then the least value of  $x^2 + ax + b (x \in R)$  is

1)  $\frac{9}{4}$  2)  $-\frac{9}{4}$  3)  $-\frac{1}{4}$  4)  $\frac{1}{4}$

48. If  $x^2 + \frac{1}{x^2} = A$  and  $x - \frac{1}{x} = B (|x| > 1)$  then

least value of  $\frac{A}{B}$  is

1) 2 2)  $\sqrt{2}$  3)  $-\sqrt{2}$  4)  $2\sqrt{2}$

49. If  $\alpha, \beta$  are the roots of  $x^2 - (a-2)x - (a+1) = 0$  where 'a' is a variable then the least value of  $\alpha^2 + \beta^2$  is

1) 2 2) 3 3) 5 4) 7

MODULUS FUNCTIONS

50. The minimum value of

$|x| + \left|x + \frac{1}{2}\right| + |x-3| + \left|x - \frac{5}{2}\right|$  is

1) 2 2) 4 3) 6 4) 4

51. Sum of the roots of the equation

$(x-2)^2 - 2|x-2| - 15 = 0$  is

1) 4 2) 0 3) -4 4) 8

52. The real roots of the equation

$|x^2 + 4x + 3| + 2x + 5 = 0$  are

1) 4;  $-1 + \sqrt{3}$  2) -4;  $-1 - \sqrt{3}$   
3) -6, -1 4) 6, -1

53. If  $x$  satisfies  $|x-1| + |x-2| + |x-3| \geq 6$  then

1)  $0 \leq x \leq 4$  2)  $x \leq -2$  (or)  $x \geq 4$   
3)  $x \leq 0$  (or)  $x \geq 4$  4)  $R$

MISCELLANEOUS

54. If  $a = \cos \frac{2\pi}{7} + i \sin \frac{2\pi}{7}$ ,  $\alpha = a + a^2 + a^4$  and  $\beta = a^3 + a^5 + a^6$  then  $\alpha, \beta$  are roots of The equation

1)  $x^2 + x + 1 = 0$  2)  $x^2 + x + 2 = 0$   
3)  $x^2 + 2x + 2 = 0$  4)  $x^2 + 2x + 3 = 0$

55. The set of values of  $x$  for which the inequality  $[x]^2 - 5[x] + 6 \leq 0$  (where  $[.]$  denote the greatest integral function) hold good is

- 1)  $2 \leq [x] < 3$       2)  $2 \leq x < 4$   
3)  $2 \leq x < 3$       4)  $2 \leq x \leq 4$

56. The number of quadratic equations which are unchanged by squaring their roots is

- 1) 2      2) 4      3) 6      4) 8

57. If  $\alpha$  is a root of the equation  $4x^2 + 2x - 1 = 0$  then the other root is given by

- 1)  $-2\alpha, -1$       2)  $4\alpha^2 + \alpha - 1$   
3)  $4\alpha^3 - 3\alpha$       4)  $4\alpha^2 - 3\alpha$

58. The least value of expression

$$x^2 + 4y^2 + 9z^2 - 2x + 8y + 27z + 15 \text{ is}$$

- 1) 15      2) 5      3) 0      4)  $-\frac{41}{4}$

59. Let  $f(x) = ax^2 + bx + c$ , if

$$f(-1) < 1, f(1) > -1, f(3) < -4 \text{ and } a \neq 0 \text{ then}$$

- 1)  $a > 0$     2)  $a > 1$       3)  $a < -\frac{1}{8}$     4)  $-\frac{1}{8} < a < 0$

60. The number of distinct pairs  $(x, y)$  of real numbers that satisfy the equation

$$4x^2 + 4xy + 2y^2 - 2y + 1 = 0$$

- 1) 0      2) 1      3) 4    4) infinitely many

### EQUATIONS REDUCIBLE TO QUADRATIC

61. The number of solutions of the equation

$$x^{\frac{2}{3}} + x^{\frac{1}{3}} - 2 = 0 \text{ is}$$

- 1) 1      2) 2      3) 3      4) 4

$$y = 2 + \frac{1}{4 + \frac{1}{4 + \frac{1}{4 + \dots \infty}}}$$

62. If      then

- 1)  $y = 6$     2)  $y = 5$       3)  $y = \sqrt{6}$     4)  $y = \sqrt{5}$

### TRANSFORMED EQUATIONS

63. If the equation  $ax^2 + bx + c = 0$  is not altered when each of the coefficient is increased by the same quantity then  $x^2 + x + 1 =$

- 1) 1      2) 0      3) 3      4) 2

64. If  $\alpha$  and  $\beta$  are the roots of the equation  $ax^2 + bx + c = 0$  and if  $px^2 + qx + r = 0$  has

roots  $\frac{1-\alpha}{\alpha}$  and  $\frac{1-\beta}{\beta}$  then  $r =$  (EAM - 2007)

- 1)  $a + 2b$     2)  $a + b + c$     3)  $ab + bc + ca$     4)  $abc$

### RANGE

65. Given that, for all real  $x$ , the expression

$$\frac{x^2 - 2x + 4}{x^2 + 2x + 4} \text{ lies between } \frac{1}{3} \text{ and } 3. \text{ The values}$$

between which the expression  $\frac{9.3^{2x} + 6.3^x + 4}{9.3^{2x} - 6.3^x + 4}$

lies are

- 1)  $\frac{1}{3}$  and 3      2) -2 and 0  
3) -1 and 1      4) 0 and 2

66. Let  $f(x) = (1 + b^2)x^2 + 2bx + 1$  and  $m(b)$  be the minimum value of  $f(x)$ . As  $b$  varies, the range of  $m(b)$  is

- 1)  $[0, 1]$     2)  $\left[0, \frac{1}{2}\right]$     3)  $\left[\frac{1}{2}, 1\right]$     4)  $(0, 1]$

### INEQUALITIES

67. If  $x$  is real and  $\frac{x-1}{4x+5} < \frac{x-3}{4x-3}$  then  $x$  lies in the interval

- 1)  $\left(-\frac{3}{4}, \frac{5}{4}\right)$       2)  $\left(-\frac{5}{4}, -\frac{3}{4}\right)$   
3)  $\left(-\frac{5}{4}, \frac{3}{4}\right)$       4)  $\left(-\frac{3}{4}, \frac{7}{5}\right)$

68. If  $(\log_5 x)^2 + \log_5 x < 2$  then  $x$  belongs to the interval

- 1)  $\left(\frac{1}{25}, 5\right)$     2)  $\left(\frac{1}{5}, \frac{1}{\sqrt{5}}\right)$     3)  $(1, \infty)$     4)  $(5, 25)$

69. If  $(x-1)(x-2)(x+5) < 0$  then

- 1)  $x < -3; 0 < x < 2$     2)  $x < -5; 1 < x < 2$   
3)  $x > 2; -5 < x < 1$     4)  $x < 1; -5 < x < 2$

OTHER MODELS

70. If  $x^3 + ax + 1 = 0$  and  $x^4 + ax^2 + 1 = 0$  have common root then the exhaustive set of values of  $a$  is (EAM - 2011)

- 1)  $(-\infty, -2)$  2)  $[-2, \infty)$  3)  $\{-2\}$  4)  $[-2, 2]$

71. If  $x$  is real then the value of  $\frac{x^2 - 3x + 4}{x^2 + 3x + 4}$  lies in the interval (EAM - 2011)

- 1)  $\left[\frac{1}{3}, 3\right]$  2)  $\left[\frac{1}{5}, 5\right]$  3)  $\left[\frac{1}{6}, 6\right]$  4)  $\left[\frac{1}{7}, 7\right]$

72. In  $\Delta PQR$ ,  $\angle R = \frac{\pi}{4}$ ,  $\tan\left(\frac{P}{3}\right)$ ,  $\tan\left(\frac{Q}{3}\right)$  are the roots of the equation  $ax^2 + bx + c = 0$ , then (EAM - 2012)

- 1)  $a + b = c$  2)  $b + c = 0$   
3)  $a + c = 0$  4)  $b = c$

73. If the harmonic mean between the roots of  $(5 + \sqrt{2})x^2 - bx + (8 + 2\sqrt{5}) = 0$  is 4, then the value of  $b$  is (EAM - 2013)

- 1) 2 2) 3 3)  $4 - \sqrt{5}$  4)  $4 + \sqrt{5}$

LEVEL-II (C.W.) - KEY

- |       |       |       |       |       |       |
|-------|-------|-------|-------|-------|-------|
| 1) 1  | 2) 4  | 3) 2  | 4) 4  | 5) 1  | 6) 2  |
| 7) 3  | 8) 3  | 9) 3  | 10) 2 | 11) 4 | 12) 3 |
| 13) 4 | 14) 4 | 15) 1 | 16) 2 | 17) 2 | 18) 2 |
| 19) 2 | 20) 3 | 21) 3 | 22) 2 | 23) 3 | 24) 3 |
| 25) 4 | 26) 1 | 27) 3 | 28) 3 | 29) 3 | 30) 1 |
| 31) 3 | 32) 1 | 33) 4 | 34) 2 | 35) 2 | 36) 3 |
| 37) 3 | 38) 4 | 39) 2 | 40) 1 | 41) 2 | 42) 1 |
| 43) 3 | 44) 2 | 45) 1 | 46) 4 | 47) 2 | 48) 4 |
| 49) 3 | 50) 3 | 51) 1 | 52) 2 | 53) 3 | 54) 2 |
| 55) 2 | 56) 2 | 57) 3 | 58) 4 | 59) 3 | 60) 2 |
| 61) 2 | 62) 4 | 63) 2 | 64) 2 | 65) 1 | 66) 4 |
| 67) 3 | 68) 1 | 69) 2 | 70) 3 | 71) 4 | 72) 1 |
| 73) 4 |       |       |       |       |       |

LEVEL-II (C.W.) - HINTS

1.  $\left(\frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right)^2 = \frac{(\alpha - \beta)^2 (\alpha + \beta)^2}{(\alpha\beta)^2}$

2.  $\sqrt{\alpha\beta} = 4$ ,  $\alpha + \beta = 18$

$\therefore$  the required equation is  $x^2 - 18x + 16 = 0$

3. Roots of  $x^2 - 2x + 3 = 0$  are  $\alpha, \beta$   
The required quadratic equation is

$$x^2 - \left(\frac{\alpha^2 + \beta^2}{\alpha\beta}\right)x + 1 = 0 \Rightarrow 3x^2 + 2x + 3 = 0$$

4.  $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta)$   
 $\Rightarrow \alpha\beta = -8$

5.  $\alpha + \beta = \frac{1}{\alpha^2} + \frac{1}{\beta^2}$

6.  $\alpha = \tan 22^\circ$ ,  $\beta = \tan 23^\circ$   
 $\tan 45^\circ = \frac{\tan 22^\circ + \tan 23^\circ}{1 - \tan 22^\circ \tan 23^\circ}$

7.  $|\alpha^2 - \beta^2| = 24$

$$\frac{|b|}{a^2} \sqrt{b^2 - 4ac} = 24; 36 - 4q = 16; q = 5$$

8.  $\alpha + \beta = a + b$ ,  $\alpha\beta = ab - c \Rightarrow ab = \alpha\beta + c$

9.  $\alpha + \beta = 15$ ,  $\alpha\beta = 1$

$$\left(\frac{1}{\alpha} - 15\right)^{-2} + \left(\frac{1}{\beta} - 15\right)^{-2}; \frac{1}{\alpha} = \beta, \frac{1}{\beta} = \alpha$$

$$= \frac{1}{\alpha^2} + \frac{1}{\beta^2} = (\alpha + \beta)^2 - 2$$

10. Let equation of line be  $\frac{x}{a} + \frac{y}{b} = 1$  passing through  $(1, 1) \Rightarrow a + b = ab \dots (1)$

$$\text{Area of the triangle} = \frac{1}{2} ab = A$$

$$\therefore ab = 2A, a + b = 2A$$

11. sub  $x = 4$  in first then  $p = -7$  and  $q = \frac{p^2}{4}$

12. Product  $= 2e^{2 \log k} - 1 = 7; 2k^2 - 1 = 7$

13.  $\frac{(m+n)^2}{mn} = \frac{b^2}{ac}$  where  $m : n = 1 : 2$

14. Let original eq be  $x^2 + ax + b = 0$  from the given  $-a = 5$ ,  $b = -6$

15.  $x^2 - (A + H)x + AH = 0$

$$\text{put } H = \frac{G^2}{A} (AH = G^2)$$

16.  $\alpha + \alpha^2 = -b/a; \alpha^3 = c/a$

$$\alpha^3 + (\alpha^3)^2 + 3\alpha^3(-b/a) = \frac{-b^3}{c^3};$$

$$\frac{c}{a} + \frac{c^2}{a^2} - \frac{3bc}{a^2} = \frac{-b^3}{a^3}; a^2c + ac^2 = 3abc - b^3$$

17. Let  $\alpha = 2k - 1, \beta = 2k + 1, k \in N$

$$\alpha + \beta = -b/a = 4k > 0 \Rightarrow b < 0 \text{ and}$$

$$-b = 4ak \geq 4a \Rightarrow |b| \geq 4a$$

18. LHS always positive RHS always negative

$$LHS \neq RHS$$

19.  $b^2 - 4c = 1$

20.  $\frac{1}{\sqrt{3}} + 2 - \sqrt{3} = -p; \frac{2}{\sqrt{3}} - 1 = q; 2 + q - p = 3$

21.  $\Delta = 4(a^2 + c^2 - ac) = 2(a^2 + c^2 + (a - c)^2) > 0$

22.  $(ax + d)^2 + (bx + c)^2 = 0$

$$ax + d = 0 \text{ and } bx + c = 0$$

$$\frac{-d}{a} = \frac{-c}{b} = x \Rightarrow ac = bd \Rightarrow a^2c^2 = (bd)^2$$

23.  $3x^2 - 2(a + b + c)x + (ab + bc + ca) = 0$

$$B^2 - 4AC = 4[(a + b + c)^2 - 3(ab + bc + ca)]$$

$$= 4[a^2 + b^2 + c^2 - ab - bc - ca] \geq 0$$

$$\therefore \text{ roots are equal.}$$

$$\Delta = 4b^2 - 4ac = (a + c)^2 - 4ac$$

24.  $= (a - c)^2 \geq 0 \quad (\because 2b = a + c)$

25.  $\Delta < 0 \Rightarrow (2a + 3)^2 - 4a(3 + 5a) < 0$

$$\Rightarrow 4a^2 + 12a + 9 - 12a - 20a^2 < 0$$

$$\therefore a \in \left(-\infty, -\frac{3}{4}\right) \cup \left(\frac{3}{4}, \infty\right)$$

26.  $\Delta = 0 \Rightarrow a^2 + b^2 + c^2 - ab - bc - ca = 0$

$$\Rightarrow (a + b\omega + c\omega^2)(a + b\omega^2 + c\omega) = 0$$

$$(\text{or}) (a - b)^2 + (b - c)^2 + (c - a)^2 = 0$$

$$\Rightarrow a = b = c$$

27. If the roots are changed by same quantity, then the difference of the roots will not be change.

28. As  $bx^2 + cx + a = 0$  has imaginary roots So,  $c^2 < 4ab$  Now,  $3b^2x^2 + 6bcx + 2c^2$

$$= 3(bx + c)^2 - c^2 \geq -c^2 > -4ab$$

29.  $f(1) = 2 \Rightarrow f(1) + f(x) = f(1)f(x)$

$$\Rightarrow \therefore 2 + x^2 - 3x + 4 = 2(x^2 - 3x + 4)$$

$$\Rightarrow x^2 - 3x + 2 = 0 \Rightarrow x = 1, x = 2$$

30. **CASE I:** When  $x < 0$ ,  $\text{sgn } x = -1$ ,

$$\Rightarrow x^2 - 5x + 6 = 0 \Rightarrow x = 2, 3$$

But  $x < 0$ . Hence the equation has no solution in this case.

**CASE II:** When  $x > 0$ ,  $\text{sgn } x = 1$

$$\Rightarrow x^2 - 5x - 6 = 0 \Rightarrow x = -1, 6$$

$$\Rightarrow x = 6 [\because x > 0]$$

Hence the given equation has only one solution.

31. Verify from options

32.  $\log_2(3 - x)(1 - x) = 3 \Rightarrow (3 - x)(1 - x) = 8$

$$\Rightarrow x = 5 \text{ \& } x = -1, x = 5 \text{ fail } \therefore x = -1$$

33.  $\sin(2 \cdot 2^x) = \frac{2^x + 2^{-x}}{2} \geq 1$  and  $\sin(2 \cdot 2^x) \neq 1$

34. Let  $e^{\sin x} = t \Rightarrow t^2 - 4t - 1 = 0$

35. Let roots be  $p + iq, p - iq$

$$\text{sum of roots} = 2p = (1 - a)i \Rightarrow 1 - a = 0$$

$$a = 1 (\because p \text{ is real})$$

36.  $a\alpha^2 + b\alpha + c = 0$  and

$$b\alpha^2 + c\alpha + a = 0 \Rightarrow \alpha = 1 \therefore a + b + c = 0$$

37.  $\frac{a}{1} = \frac{b}{3} = \frac{c}{5} = \lambda (\text{say}) \Rightarrow a + b + c = 9\lambda$

38.  $\Delta = -116 < 0$  Roots are imaginary and conjugate to each other. Both roots are common

$$\therefore \frac{a}{3} = \frac{2b}{8} = \frac{3c}{15}$$

$$a = 3, b = 4, c = 5$$

$$\Delta ABC \text{ is right angle triangle } c^2 = a^2 + b^2$$

39.  $af(k) < 0$  and  $\Delta > 0$



40. The roots of  $x^2 - 2mx + m^2 - 1 = 0$  are

$$\frac{2m \pm \sqrt{4m^2 - 4(m^2 - 1)}}{2} = m \pm 1$$

Given  $-2 < m - 1, m + 1 < 4 \Rightarrow m > -1, m < 3$

41.  $(x-1)^2 = a^2 \Rightarrow x = 1 \pm a$

let  $f(x) = x^2 - 2(a+1)x + a(a-1)$

then  $f(1+a) < 0$  and  $f(1-a) < 0$

42.  $|\alpha - \beta| < \sqrt{5} \Rightarrow \sqrt{a^2 - 4} < \sqrt{5}$

43.  $(x+11)(x-7) < 0$  and  $(x+2)(x-2) > 0$

$$x \in (-11, -2) \cup (2, 7)$$

44. Let  $3^x = t \quad \therefore 3^x + 3^{1-x} - 4 < 0$

$$\Rightarrow t^2 - 4t + 3 < 0 \Rightarrow (t-1)(t-3) < 0$$

$$\therefore 1 < t < 3 \Rightarrow 1 < 3^x < 3 \Rightarrow x \in (0, 1)$$

45.  $4a + 4b - 3c > 0 \Rightarrow f(2) > 0$

$$\therefore f(x) > 0 \forall x \in \mathbb{R} \quad \therefore f(0) > 0$$

46.  $\Delta = \cos^2 p - 4(\cos p - 1)(\sin p)$

$$= (\cos p - 2 \sin p)^2 + 4(1 - \sin p) \sin p$$

$$\Delta \geq 0 \Rightarrow \sin p \geq 0 \Rightarrow p \in (0, \pi)$$

47.  $a + b = -a, ab = b \Rightarrow a = 1, b = -2$

$$\min = \frac{4b - a^2}{4}$$

$$48. \frac{A}{B} = \frac{x^2 + \frac{1}{x^2}}{x - \frac{1}{x}} = \frac{\left(x - \frac{1}{x}\right)^2 + 2}{x - \frac{1}{x}}$$

$$= \left(x - \frac{1}{x}\right) + \frac{2}{x - \frac{1}{x}} \text{ where } p = x - \frac{1}{x}$$

$$= p + \frac{2}{p} \geq 2\sqrt{2} \quad (A.M \geq G.M)$$

49.  $\alpha^2 + \beta^2 = (a-2)^2 + 2(a+1)$

$$= (a-1)^2 + 5 \geq 5$$

50. The minimum of  $\left\{f(0), f\left(\frac{-1}{2}\right), f(3), f\left(\frac{5}{2}\right)\right\}$

51.  $(|x-2|-5)(|x-2|+3) = 0$

$$|x-2| = 5, |x-2| + 3 \neq 0$$

$$x = \pm 5 + 2 \Rightarrow x = 7, -3$$

52. -4 satisfies the equation

53. suppose  $x \geq 3$  eq becomes

$$(x-1) + (x-2) + (x-3) \geq 6 \Rightarrow x \geq 4$$

suppose  $2 \leq x \leq 3$  eq becomes

$$(x-1) + (x-2) - (x-3) \geq 6 \Rightarrow x \geq 6$$

suppose  $1 \leq x \leq 2$  eq becomes

$$(x-1) + (x-2) - (x-3) \geq 6 \Rightarrow x \leq -2$$

suppose  $x \leq 1$  eq becomes

$$-(x-1) - (x-2) - (x-3) \geq 6 \Rightarrow x \leq 0$$

from the above cases  $x \leq 0$  or  $x \geq 4$

54.  $\alpha = a + a^2 + a^4, \beta = a^3 + a^5 + a^6$

$1, a, a^2, a^3, a^4, a^5, a^6$  are 7th roots of unity.

$$1 + a + a^2 + a^3 + a^4 + a^5 + a^6 = 0$$

$$\alpha + \beta = -1$$

$$\alpha\beta = 3 + a + a^2 + a^3 + a^4 + a^5 + a^6 = 2$$

55.  $[x^2] - 5[x] + 6 \leq 0; ([x] - 2)([x] - 3) \leq 0$

$$2 \leq [x] \leq 3; 2 \leq x < 4$$

56.  $x^2 = 0, x^2 - x = 0, x^2 + x + 1 = 0, x^2 - 2x + 1 = 0$

57.  $4x^2 + 2x - 1 = 0 \quad \alpha = \frac{\sqrt{5}-1}{4} = \sin 18^\circ;$

$$\beta = -\left(\frac{\sqrt{5}+1}{4}\right) = -\sin 54^\circ$$

$$= -(3\alpha - 4\alpha^3) = 4\alpha^3 - 3\alpha$$

58.  $x^2 + 4y^2 + 9z^2 - 2x + 8y + 27z + 15$

$$= (x-1)^2 + 4(y+1)^2 + 9\left(z + \frac{3}{2}\right)^2 - \frac{41}{4} \geq -41/4$$

59.  $a - b + c < 1, a + b + c > -1; 9a + 3b + c < -4$

$$\Rightarrow 12a + 4c < -1 \text{ and}$$

$$6a - 2c < -1 \Rightarrow 24a < -3$$

60. The equation is  $(2x+y)^2 + (y-1) = 0$

